- Introduction
- Acid – base equilibria
- Calculation of pH
- Strong acid – base titration
- Weak acid – base titration
- Polyprotic acid – base titration

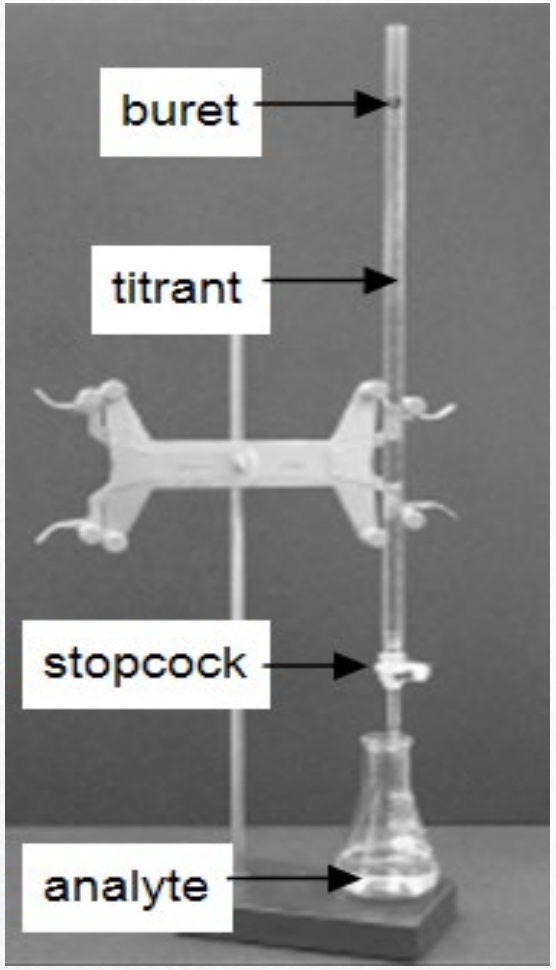


3 Acid – Base Titration

- Indicator and titration error
- Application

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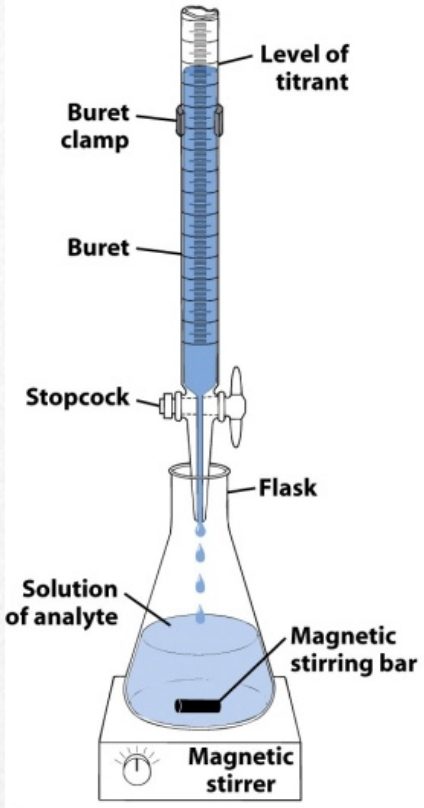
1. Introduction



1. Introduction

- Titration is a technique used to determine the concentration of solution.
- It is one of the most important method used in quantitative analysis.
- Requires:
  - Careful manipulation
  - Precise buret reading

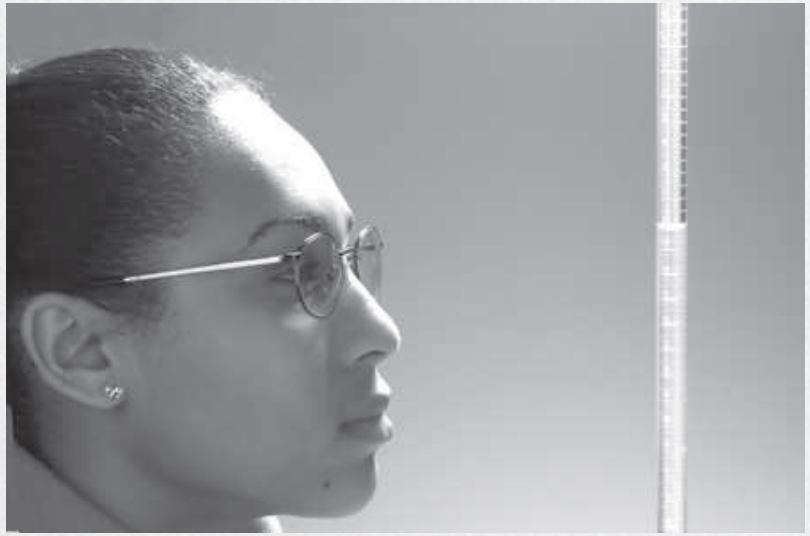
Read from the bottom of the meniscus







12.67 mL





12.58 mL

√

12.61 mL

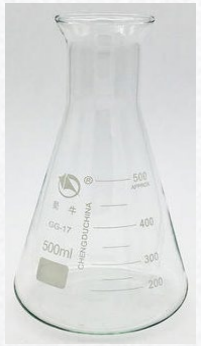




To avoid the problem of parallax, buret readings should be made consistently along a line perpendicular to the buret

1.1 Volume Measuring Implement





Conical flask



Volumetric flask



beaker



Measuring cylinder



volumetric pipette

1.1 Volume Measuring Implement

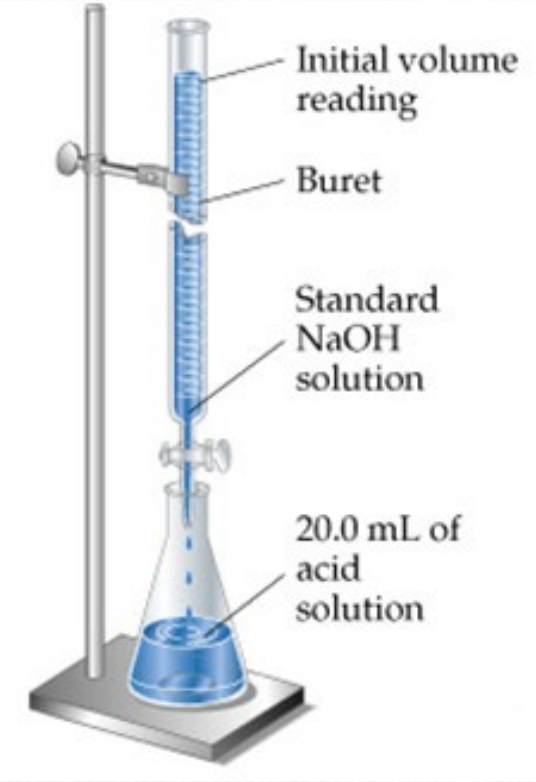
Manufacturers' tolerances on some items of glassware

Item of glassware	Grade A	Grade B
1 ml bulb pipette	± 0.7%	± 1.5%
5 ml bulb pipette	± 0.3%	± 0.6%
100 ml volumetric flask	± 0.08%	± 0.15%
500 ml volumetric flask	± 0.05%	± 0.1%
Full 25 ml burette	± 0.2%	± 0.4%

1.2 Relevant terms in titration

- **Standard solution (Titrant)**
- Equivalence point
- Indicator
- End point
- Titration error

A solution used to determine the concentration of an analyte. The **accurate concentration** of the solution **is known**.



1.2 Relevant terms in titration

- Standard solution (Titrant)
- **Equivalence point**
- Indicator
- End point
- Titration error

is a theoretical point reached when the amount of added titrant is chemically equivalent to the amount of analyte in the sample.

$$H_2SO_4 + \mathbf{2NaOH} \longrightarrow Na_2SO_4 + \mathbf{2H_2O}$$

1.2 Relevant terms in titration

- Standard solution (Titrant)
- Equivalence point
- **Indicator**
- End point
- Titration error

Indicators are often added to the analyte solution to produce an **observable physical change** (signaling the end point) at or near the equivalence point.

$$H_2SO_4 + \mathbf{2NaOH} \longrightarrow Na_2SO_4 + \mathbf{2H_2O}$$

1.2 Relevant terms in titration

- Standard solution (Titrant)
- Equivalence point
- Indicator
- **End point**
- Titration error

The position of some physical change (color change of the solution or potential of an electrode) associated with the condition of chemical equivalence is called the end point for the titration.

$$H_2SO_4 + \mathbf{2NaOH} \longrightarrow Na_2SO_4 + \mathbf{2H_2O}$$

1.2 Relevant terms in titration

- Standard solution (Titrant)
- Equivalence point
- Indicator
- End point
- **Titration error**

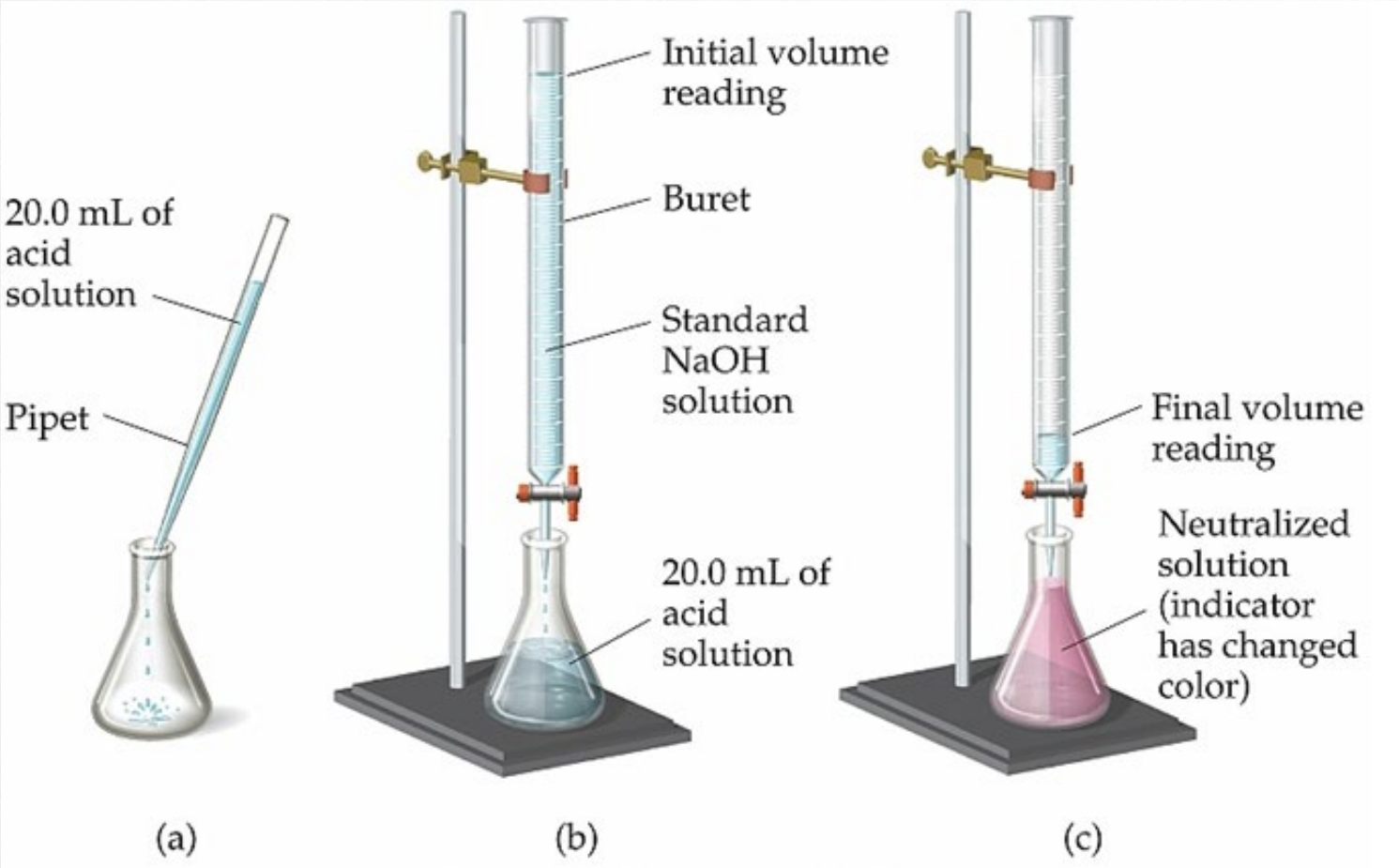
- The difference between the end point and the equivalent point.
- Unsuitable indicator will cause large titration error.

$$H_2SO_4 + \mathbf{2NaOH} \longrightarrow Na_2SO_4 + \mathbf{2H_2O}$$

2. Acid – base equilibria

- The **neutralization** of an acid with a base.
- It is used to **quantify** the **amount** of the **unknown acid or base**.
- Acid + Base → Water + Salt
- Ex: HCl + NaOH → H_2O + NaCl

2. Acid – base equilibria



$HCl + NaOH \rightarrow H_2O + NaCl$

$$c_{HCl} = \frac{c_{NaOH} \times V_{NaOH}}{V_{HCl}}$$

2.1 Definition of acid and base

- **Acid**: a substance that can give up protons (**proton donor**)

$$HA \rightarrow H^+ + A^-$$ **acid**

- **Base**: any compound or ion that can accept protons (**proton acceptor**).

$$A^- + H_2O \rightarrow HA + OH^-$$ **base**

- HA and A^- is **conjugate acid-base pair**.

- When an acid donate a proton, it becomes its **conjugated** base.

$$\boldsymbol{HA} \rightarrow H^+ + \boldsymbol{A^-}$$

acid **base**

2.1 Definition of acid and base

- It can be an acid, also can be a base.
  - For example, HCO_3^-

Acid		Base		Proton
HAC	⇌	Ac^-	+	H^+
H_2CO_3	⇌	HCO_3^-	+	H^+
HCO_3^-	⇌	CO_3^{2-}	+	H^+
NH_4^+	⇌	NH_3	+	H^+
H_6Y^{2+}	⇌	H_5Y^+	+	H^+
NH_3OH^+	⇌	NH_2OH	+	H^+
$(CH_2)_6N_4H^+$	⇌	$(CH_2)_6N_4$	+	H^+

2.2 Essential of acid – base reaction

- **Half reaction** of acid – base： the acid donate proton and the base accept proton.
  - – Dissociation of acetic acid in water
    - Half reaction 1
    - Half reaction 2

$$HAc\ \ (acid_1) \Leftrightarrow Ac^-(base_1) + H^+$$

$$H^+ + H_2O\ \ (base_2) \Leftrightarrow H_3O^+(acid_2)$$

$$HAc\ \ (acid_1) + H_2O\ \ (base_2) \Leftrightarrow H_3O^+(acid_2) + Ac^-(base_1)$$

Conjugated acid – base pair

2.2 Essential of acid – base reaction

- **Half reaction** of acid – base： the acid donate proton and the base accept proton.
  - Dissociation of ammonia in water
    - Half reaction 1
    - Half reaction 2

$$NH_3(base_1) + H^+ \Leftrightarrow NH_4^+(acid_1)$$

$$H_2O\ (acid_2) \Leftrightarrow OH^-(base_2) + H^+$$

$$NH_3(base_1) + H_2O\ (acid_2) \Leftrightarrow OH^-(base_2) + NH_4^+(acid_1)$$

Conjugated acid – base pair

2.2 Essential of acid – base reaction

- Dissociation of NH_4Cl（dissociation of weak acid NH_4^+）

$$\underbrace{NH_4^+ + H_2O \Leftrightarrow H_3O^+ + NH_3}$$

Conjugated acid – base pair

- Dissociation of NaAc（dissociation of weak base Ac^-）

$$\underbrace{Ac^- + H_2O \Leftrightarrow OH^- + HAc}$$

Conjugated acid – base pair

2.2 Essential of acid – base reaction

- Half reaction can **NOT** happen alone
- Acid – base reaction is the result of two sets of conjugated acid – base pair
- Essential of acid – base reaction is the **transfer of proton**.

2.3 Dissociation of water and ion – product constant K_w

- **Autoprotolysis**: it is *self – ionization* of **amphiprotic** solvents
- H_2O can be a proton donor, also can be a proton acceptor.
- In aqueous solution

$$H_2O + H_2O \Leftrightarrow H_3O^+ + OH^- \qquad K_s = K_w = [H^+][OH^-] = 1.0 \times 10^{-14}$$

- Non – aqueous solution

$$SH + SH \Leftrightarrow SH_2^+ + S^- \qquad K_s = [SH_2^+][S^-]$$

2.4 Strengths of Acids and Bases

- **Strong acid:** Reaction with the solvent is sufficiently **complete** that **NO undissociated solute molecules** are left in aqueous solution.
- **Weak acid:** React **incompletely** with water to give solutions **containing** significant quantities of both **the parent acid** and **its conjugated bases**.

  - **Note: acid can be cationic, anionic or electrically neutral.**
  - **The same holds for bases.**

2.5 Dissociation constant K_a and K_b

- The dissociation constant of weak acid K_a:
- **Weak acid:** $HA \Leftrightarrow A^- + H^+$

$$K_a = \frac{[H^+][A^-]}{[HA]}$$

The acid with larger K_a value indicates stronger acidity.

2.5 Dissociation constant K_a and K_b

- The dissociation constant of weak base K_b:
- **Weak base**: $A^- + H_2O \Leftrightarrow HA + OH^-$

$$K_b = \frac{[HA][OH^-]}{[A^-]}$$

The base with larger K_b value indicates stronger basicity.

2.6 Dissociation constant for conjugated acid/base pair

- In water, the acid and the base constitute conjugate acid-base pair.
- Example:

$$NH_3 + H_2O \overset{K_b}{\Leftrightarrow} NH_4^+ + OH^- \qquad K_b = \frac{[NH_4^+][OH^-]}{[NH_3]}$$

$NH_4Cl \rightarrow Cl^-$

$NH_4Cl \rightarrow NH_4^+$

$$NH_4^+ + H_2O \overset{K_a}{\Leftrightarrow} NH_3 + H_3O^+ \qquad K_a = \frac{[H_3O^+][NH_3]}{[NH_4^+]}$$

$$K_a \times K_b = \frac{[H_3O^+][NH_3]}{[NH_4^+]} \times \frac{[NH_4^+][OH^-]}{[NH_3]} = [H_3O^+][OH^-] = 10^{-14}$$

K_w

2.6 Dissociation constant for conjugated acid/base pair

$$K_a \times K_b = K_w = 10^{-14}$$

A strong acid forms a weak conjugated base

$$H_3PO_4 \underset{K_{b_3}}{\overset{K_{a_1}}{\Longleftrightarrow}} H^+ + H_2PO_4^- \qquad H_2PO_4^- \underset{K_{b_2}}{\overset{K_{a_2}}{\Longleftrightarrow}} H^+ + HPO_4^{2-} \qquad HPO_4^{2-} \underset{K_{b_1}}{\overset{K_{a_3}}{\Longleftrightarrow}} H^+ + PO_4^{3-}$$

$$K_{a_1} > K_{a_2} > K_{a_3} \qquad K_{b_1} > K_{b_2} > K_{b_3}$$

$$K_{a_1} \cdot K_{b_3} = K_{a_2} \cdot K_{b_2} = K_{a_3} \cdot K_{b_1} = K_w = 1.0 \times 10^{-14}$$

2.7 Fraction (δ_i)

- When a weak acid ionized in the solution:

  – $HA \Leftrightarrow H^+ + A^-$ **Non – ionized form** **Ionized form**

  – Two fractions of HA in the solution: **HA** and **A^-**

- When it reaches equilibration, the ***molar concentration*** of each ion or molecule ***in the solution*** is showed as:

  - **[HA]** and **[A^-]**.

$$c_{HA} = [HA] + [A^-]$$

2.7 Fraction (δ_i)

$$\delta_i = \frac{[i]}{c}$$

← Equilibrium molarity

← Total molarity

$$HA \Leftrightarrow H^+ + A^-$$

$$\begin{cases} \delta_{HA} = \dfrac{[HA]}{c_{HA}} \\ \delta_{A^-} = \dfrac{[A^-]}{c_{HA}} \end{cases}$$

$$c_{HA} = [HA] + [A^-]$$

$$\delta_{HA} + \delta_{A^-} = \frac{[HA]}{c_{HA}} + \frac{[A^-]}{c_{HA}} = 1$$

2.7 Fraction (δ_i)

$$HA \Leftrightarrow H^+ + A^- \qquad K_a = \frac{[H^+][A^-]}{[HA]}$$

$$\delta_0 = \delta_{HA} = \frac{[HA]}{c_{HA}} = \frac{[HA]}{[HA] + [A^-]} = \frac{1}{1 + \frac{[A^-]}{[HA]}} = \frac{[H^+]}{[H^+] + K_a}$$

$$\delta_1 = \delta_{A^-} = \frac{K_a}{[H^+] + K_a}$$

$$\delta_{HA} + \delta_{A^-} = \frac{[H^+]}{[H^+] + K_a} + \frac{K_a}{[H^+] + K_a} = 1$$

2.7 Fraction (δ_i)

Calculate the fraction ($\boldsymbol{\delta_i}$) and equilibrium concentrations of HAc and Ac^- in a **0.10 mol/L** HAc solution at **pH = 5.00**. (pKa = 4.76)

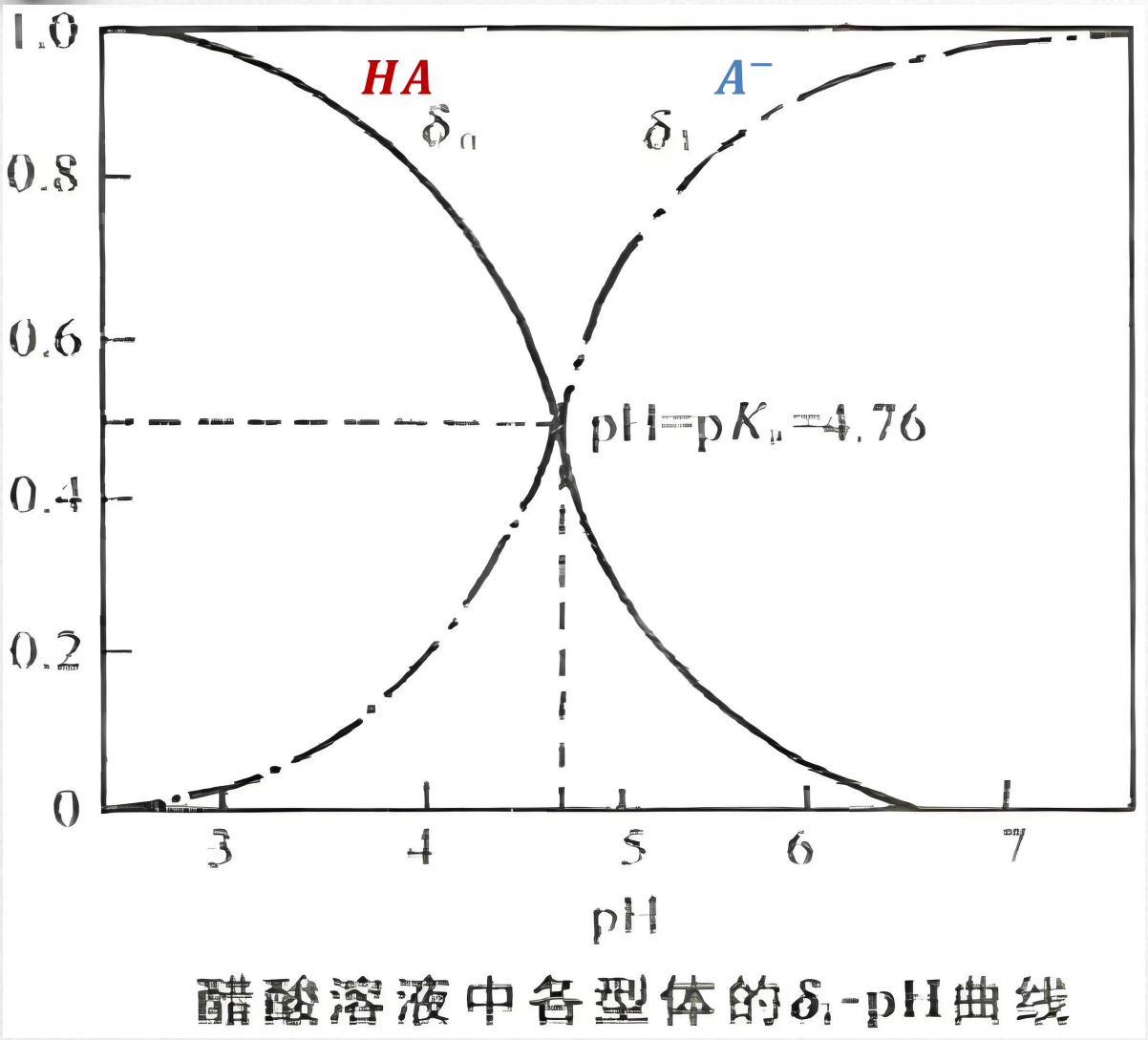
$$\boldsymbol{\delta_0 = \delta_{HAc} = \frac{[H^+]}{[H^+] + K_a} = \frac{10^{-5}}{10^{-5} + 10^{-4.76}} = 0.36}$$

$$\boldsymbol{\delta_1 = \delta_{Ac^-} = 1 - \delta_0 = 1 - 0.36 = 0.64}$$

$$\boldsymbol{[HAc] = \delta_0 \times c_{HAc} = 0.36 \times 0.10 = 0.036\ mol/L}$$

$$\boldsymbol{[Ac^-] = \delta_1 \times c_{HAc} = 0.64 \times 0.10 = 0.064\ mol/L}$$

2.7 Fraction (δ_i)



醋酸溶液中各型体的δ_i-pH曲线

$$HA \overset{K_a}{\Leftrightarrow} A^- + H^+$$

$$K_a = \frac{[H^+][A^-]}{[HA]}$$

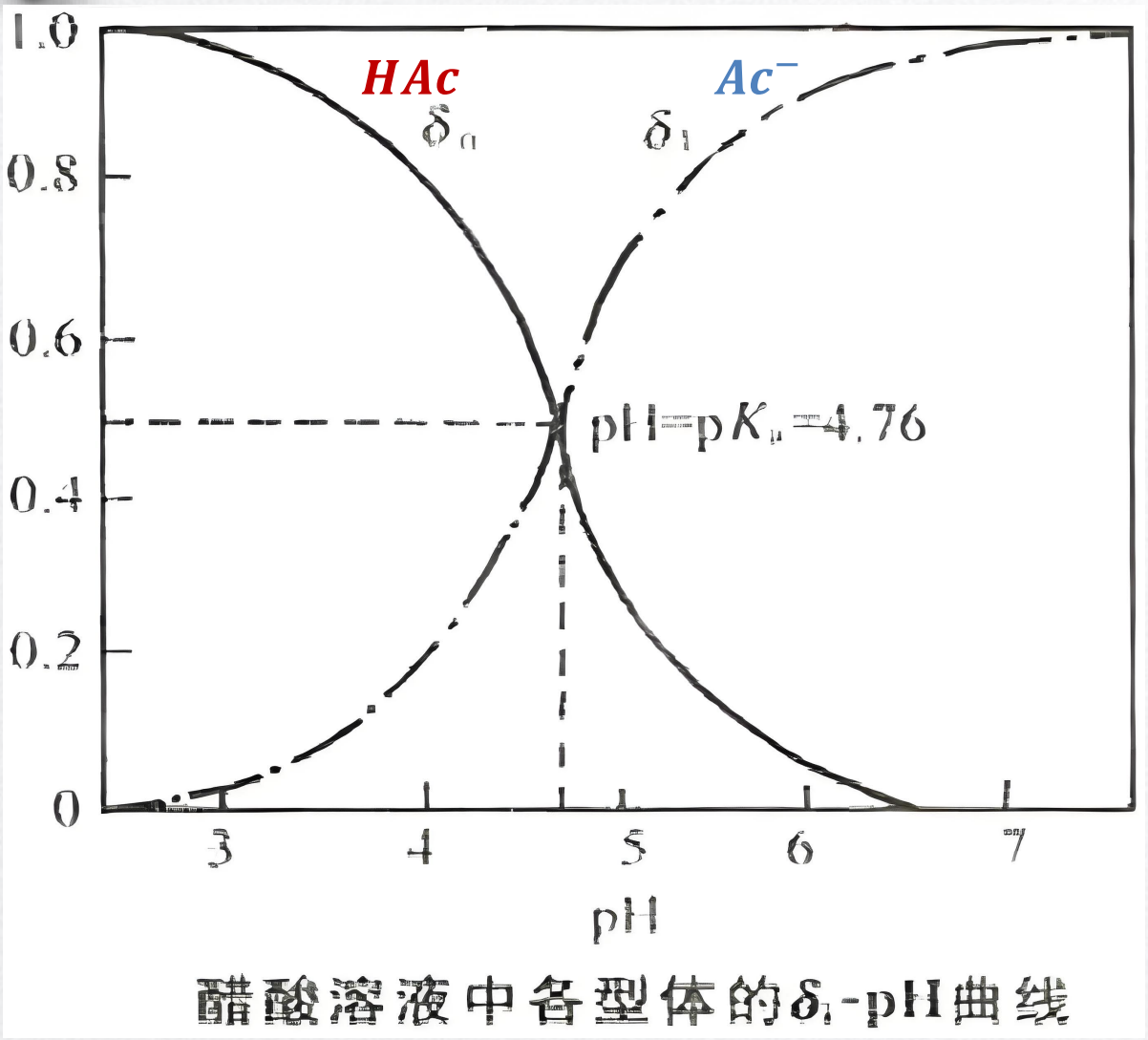
pH = pKa, $[H^+] = K_a$

$$\frac{[A^-]}{[HA]} = 1$$

50% ionization

2.7 Fraction (δ_i)

$$HAc \overset{K_a}{\Leftrightarrow} Ac^- + H^+$$



醋酸溶液中各型体的δ_i-pH曲线

$$K_a = \frac{[H^+][Ac^-]}{[HAc]} = 10^{-4.76}$$

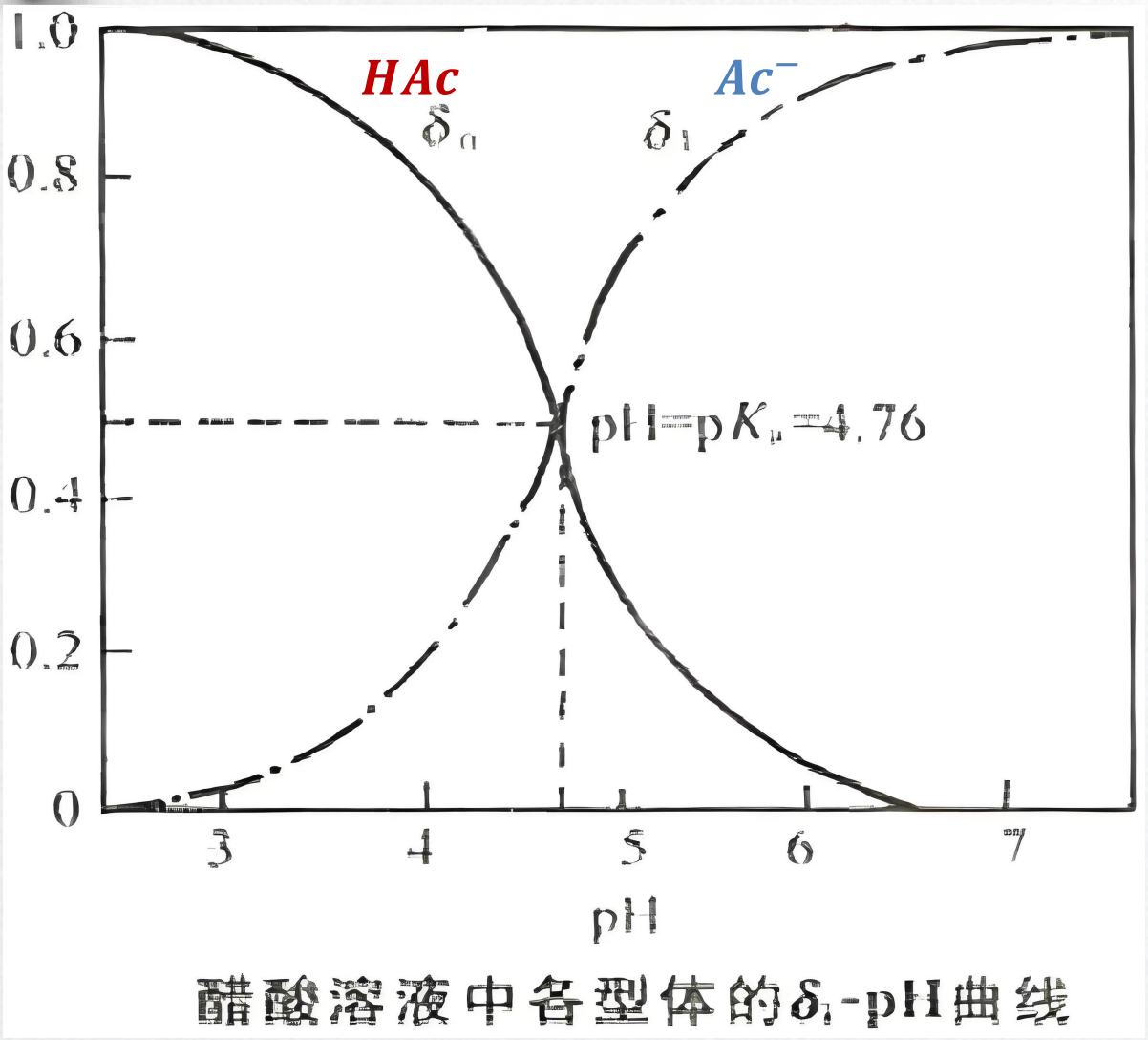
$$pK_a = p\frac{[H^+][A^-]}{[HA]}$$

$$pK_a = pH + p\frac{[A^-]}{[HA]}$$

$$pH = pK_a + lg\frac{[A^-]}{[HA]}$$

2.7 Fraction (δ_i)

$$HAc \overset{K_a}{\Leftrightarrow} Ac^- + H^+$$



醋酸溶液中各型体的δ_i-pH曲线

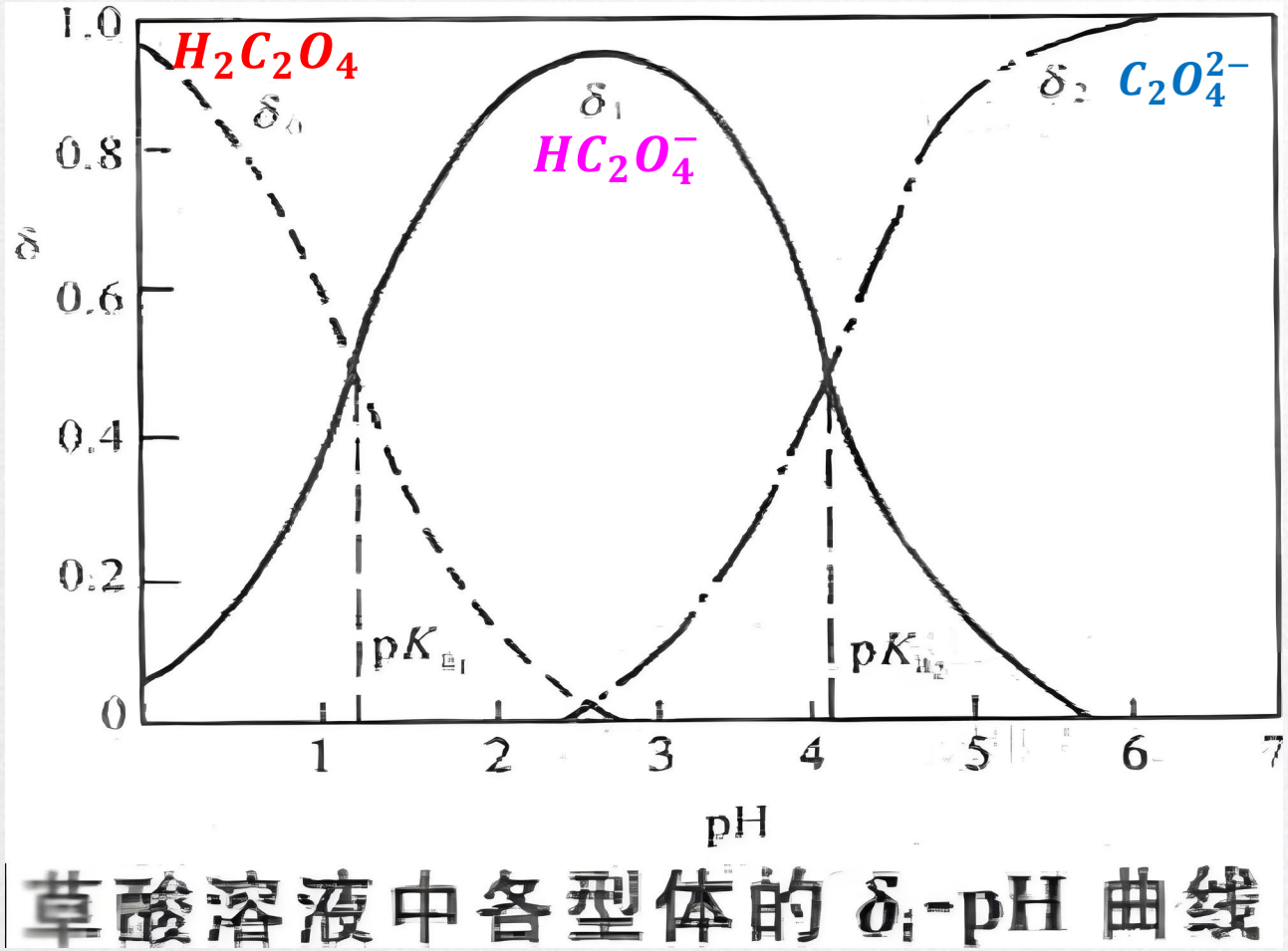
$$pH = 4.76 + lg\frac{[Ac^-]}{[HAc]}$$

When pH = 4.76, 50% of acetic acid is in ionization form (Ac^-)

Question: When pH = 5.76 or 3.76, ionization percentage of acetic acid is _____.

2.7 Fraction (δ_i)

$$H_2C_2O_4$$



草酸溶液中各型体的 δ_i-pH 曲线

$$H_2C_2O_4 \overset{K_{a_1}}{\longleftrightarrow} HC_2O_4^- + H^+$$

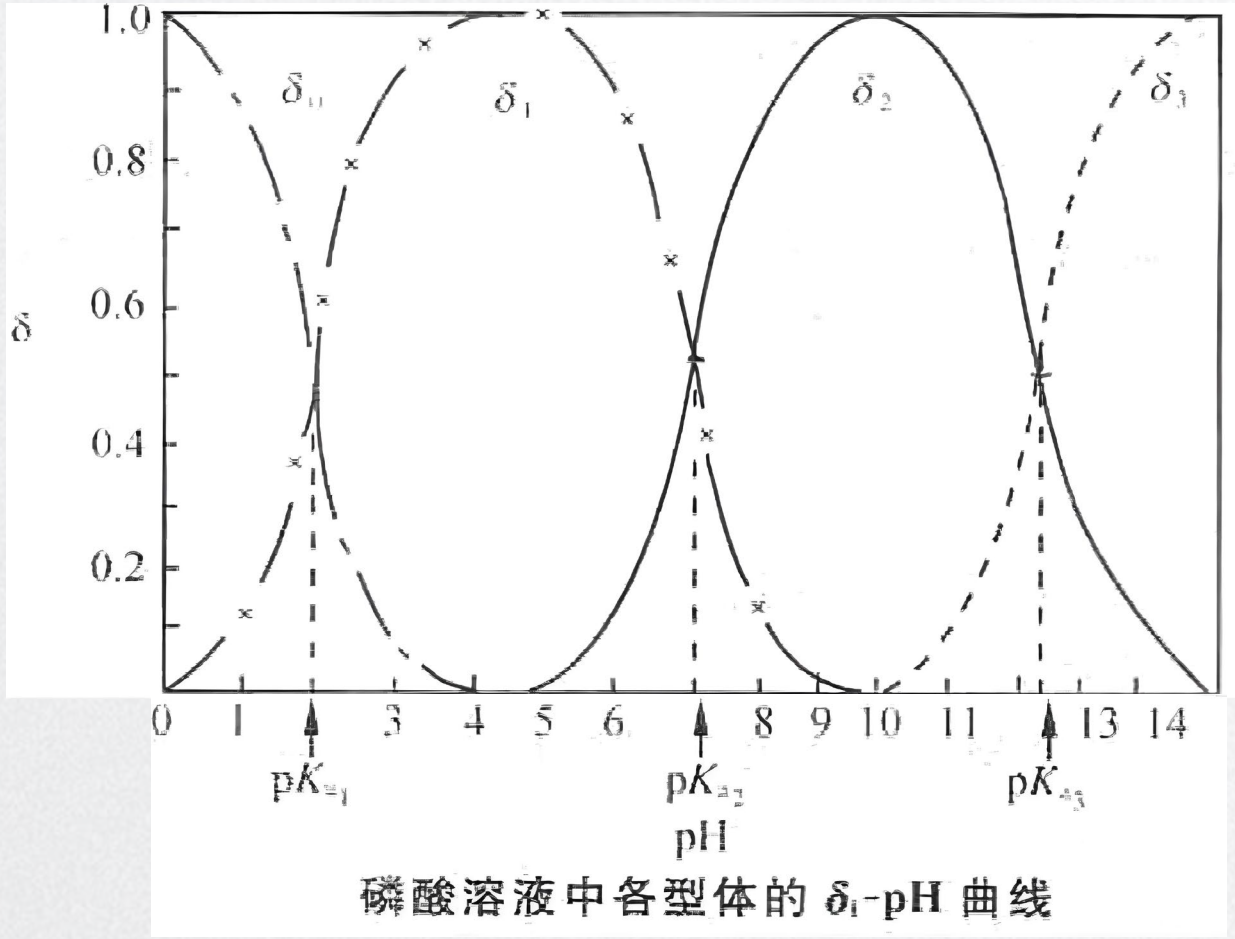
$\delta_0 \downarrow$ $\delta_1 \uparrow$

$$HC_2O_4^- \overset{K_{a_2}}{\longleftrightarrow} C_2O_4^{2-} + H^+$$

$\delta_1 \downarrow$ $\delta_2 \uparrow$

2.7 Fraction (δ_i)

$$H_3PO_4$$



磷酸溶液中各型体的 δ_i-pH 曲线

$$H_3PO_4 \overset{K_{a_1}}{\Longleftrightarrow} H^+ + H_2PO_4^-$$

$$H_2PO_4^- \overset{K_{a_2}}{\Longleftrightarrow} H^+ + HPO_4^{2-}$$

$$HPO_4^{2-} \overset{K_{a_3}}{\Longleftrightarrow} H^+ + PO_4^{3-}$$

3. Calculation of pH



01 **Strong acid and base**

02 **Weak acid and base**

03 **Polyprotic acids/bases and Amphiprotic solution**

04 **Buffer**

3. Calculation of pH

- **pH** is frequently used to express the **acidity of an aqueous solution**.
- An approximate definition of pH is the **negative logarithm of the H^+ concentration** ($[H^+]$)
  - **$pH = -\lg [H^+]$**
- pH value can be affected by the temperature.

pH of H_2O

$$K_w = [H^+][OH^-] = 1.0 \times 10^{-14}$$

- **In pure water at 25 ℃**
  - **$[H^+] = 1.0 \times 10^{-7}$**
  - **$pH = -\lg (1.0 \times 10^{-7}) = 7$**

3.1 pH calculation of strong acid and base

$$HA \leftrightarrow A^- + H^+$$

- A strong acid is *completely dissociated* in aqueous solution
- c_a (mol/L): the analytical molarity of a strong acid HA.

Ignore proton dissociated from water, then:

$$[H^+] \approx c_a \qquad \mathbf{pH} = -\mathbf{lg}[H^+] = -\mathbf{lg}c_a$$

pH of 0.10 mol/L HCl is: $\mathbf{pH} = -\mathbf{lg}\,0.1 = 1$

3.1 pH calculation of strong acid and base

- **What is the pH of 10^{-3}mol/L HCl in water?**
  - $[H^+]=10^{-3}mol/L$
  - $pH = -lg[H^+] = -lg\ 10^{-3} = 3$
- **What is the pH of 10^{-5}mol/L NaOH in water?**
  - $[OH^-]=10^{-5}mol/L$

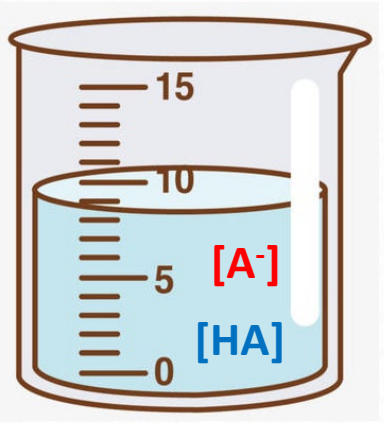
$$[H^+] = \frac{K_w}{[OH^-]} = \frac{10^{-14}}{10^{-5}} = 10^{-9} mol/L$$

  - $pH = -lg[H^+] = -lg\ 10^{-9} = 9$

3.2 pH calculation of weak acid and base

$$HA \overset{K_a}{\Leftrightarrow} H^+ + A^-$$

$$K_a = \frac{[H^+][A^-]}{[HA]}$$

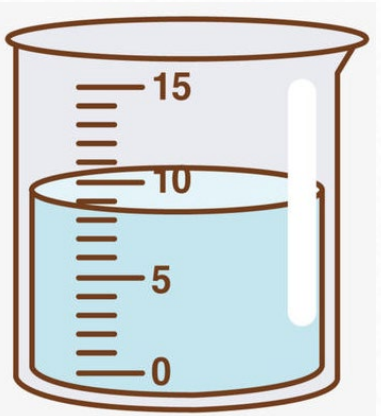


$$[H^+] = [A^-] + [OH^-]$$

$$[H^+] = c_a \cdot \delta_{A^-} + \frac{K_w}{[H^+]} = c_a \cdot \frac{K_a}{[H^+] + K_a} + \frac{K_w}{[H^+]}$$

$$[H^+]^3 + K_a[H^+]^2 - (c_a K_a + K_w)[H^+] - K_a K_w = 0$$

3.2 pH calculation of weak acid and base



$$HA \overset{K_a}{\Leftrightarrow} H^+ + A^-$$

$$K_a = \frac{[H^+][A^-]}{[HA]}$$

$$[H^+] = [A^-] + [OH^-] \Rightarrow [H^+] = \frac{K_a[HA]}{[H^+]} + \frac{K_w}{[H^+]}$$

If $c_aK_a \geq 20K_w$, ignore proton dissociated from water

$$[H^+] = \sqrt{K_a[HA]} \Leftarrow [H^+] = \frac{K_a[HA]}{[H^+]}$$

3.2 pH calculation of weak acid and base

$$HA \overset{K_a}{\Leftrightarrow} H^+ + A^-$$

$$K_a = \frac{[H^+][A^-]}{[HA]}$$

$$[H^+] = \sqrt{K_a[HA]}$$

- If $\frac{c_a}{K_a} \geq 500$, [HA] can be considered as c_a
- This is the simplified equation of weak acid.

$$[H^+] = \sqrt{K_a[HA]} = \sqrt{K_a c_a}$$

- The simplified equation of a **weak base** is $[OH^-] = \sqrt{c_b K_b}$

3.3 Calculation of pH – polyprotic acids/bases

- Acids with **more than one** acidic hydrogen ionize in steps.

Phosphate acid H_3PO_4

$$H_3PO_4 \overset{K_{a_1}}{\Longleftrightarrow} H^+ + H_2PO_4^-$$

$$H_2PO_4^- \overset{K_{a_2}}{\Longleftrightarrow} H^+ + HPO_4^{2-}$$

$$HPO_4^{2-} \overset{K_{a_3}}{\Longleftrightarrow} H^+ + PO_4^{3-}$$

Malonic acid

$$CH_2(COOH)_2 \overset{K_{a_1}}{\Longleftrightarrow} H^+ + HOOC - CH_2 - COO^-$$

$$HOOC - CH_2 - COO^- \overset{K_{a_2}}{\Longleftrightarrow} H^+ + CH_2(COO)^{2-}$$

3.3 Calculation of pH – polyprotic acids/bases

- If $\frac{K_{a_1}}{K_{a_2}} \geq 100$, the second dissociation will have little effect and can be ignored.

$$H_2A \overset{K_{a_1}}{\Longleftrightarrow} H^+ + HA^-$$

$$HA^- \overset{K_{a_2}}{\Longleftrightarrow} H^+ + A^{2-}$$

- The pH of the solution is calculated from the K_{a1} expression.

3.3 Calculation of pH – polyprotic acids/bases

- **Calculate the pH of 0.15 mol/L solution of malonic acid.**
- **$K_{a1}=10^{-2.85}$ and $K_{a2}=10^{-5.66}$.**

$$\frac{K_{a_1}}{K_{a_2}} = \frac{10^{-2.85}}{10^{-5.66}} = 10^{2.81} \geq 100$$

The calculation of pH only consider the protons dissociate from the first step.

3.3 Calculation of pH – polyprotic acids/bases

- **Calculate the pH of 0.15 mol/L solution of malonic acid.**
- **$K_{a1}=10^{-2.85}$ and $K_{a2}=10^{-5.66}$.**

Criteria

- $c_a K_{a1} \geq 20 K_w$
- $\frac{c_a}{K_{a_1}} \geq 500$

$$c_a K_{a_1} = 0.15 \times 10^{-2.85} \geq 20 \times 10^{-14}$$

But, $$\frac{c_a}{K_{a_1}} = \frac{0.15}{10^{-2.85}} = 0.15 \times 10^{2.85} = 106 < 500$$

Thus, $$[H^+] = \sqrt{K_a(c_a - [H^+])} \Rightarrow [H^+] = \sqrt{10^{-2.85} \times (0.15 - [H^+])}$$

$$[H^+] = 1.38 \times 10^{-2} \Rightarrow pH = -\lg(1.38 \times 10^{-2}) = 1.86$$

3.3 Calculation of pH – polyprotic acids/bases

Calculate pH of 0.10 mol/L $Na_2C_2O_4$ solution.

($H_2C_2O_4$, $K_{a1} = 5.6\times10^{-2}$; $K_{a2} = 5.4\times10^{-5}$)

$$C_2O_4^{2-} + H_2O \underset{K_{a2}}{\overset{K_{b_1}}{\Longleftrightarrow}} HC_2O_4^- + OH^- \qquad K_{b_1} = \frac{K_w}{K_{a_2}} = \frac{10^{-14}}{5.4\times10^{-5}} = 1.85\times10^{-10}$$

$$HC_2O_4^- + H_2O \underset{K_{a1}}{\overset{K_{b_2}}{\Longleftrightarrow}} H_2C_2O_4 + OH^- \qquad K_{b_2} = \frac{K_w}{K_{a_1}} = \frac{10^{-14}}{5.6\times10^{-2}} = 1.79\times10^{-13}$$

The calculation of pH only consider the dissociate from the first step.

$$\frac{K_{b_1}}{K_{b_2}} = \frac{1.85\times10^{-10}}{1.79\times10^{-13}} = 1.03\times10^{3} \geq 100$$

3.3 Calculation of pH – polyprotic acids/bases

Calculate pH of 0.10 mol/L $Na_2C_2O_4$ solution.

($H_2C_2O_4$, $K_{a1} = 5.6\times10^{-2}$; $K_{a2} = 5.4\times10^{-5}$)

$$c_b K_{b_1} = 0.10 \times 1.85 \times 10^{-10} = 1.85 \times 10^{-11} \geq 20 \times 10^{-14}$$

$$\frac{c_b}{K_{b_1}} = \frac{0.1}{1.85 \times 10^{-10}} = 5.4 \times 10^{8} > 500$$

$$[OH^-] = \sqrt{c_b K_{b_1}} = \sqrt{0.10 \times 1.85 \times 10^{-10}} = 4.3 \times 10^{-6}$$

$$pOH = -\lg[OH^-] = -\lg\left(4.3 \times 10^{-6}\right) = 5.37$$

$$pH = 14 - 5.37 = 8.63$$

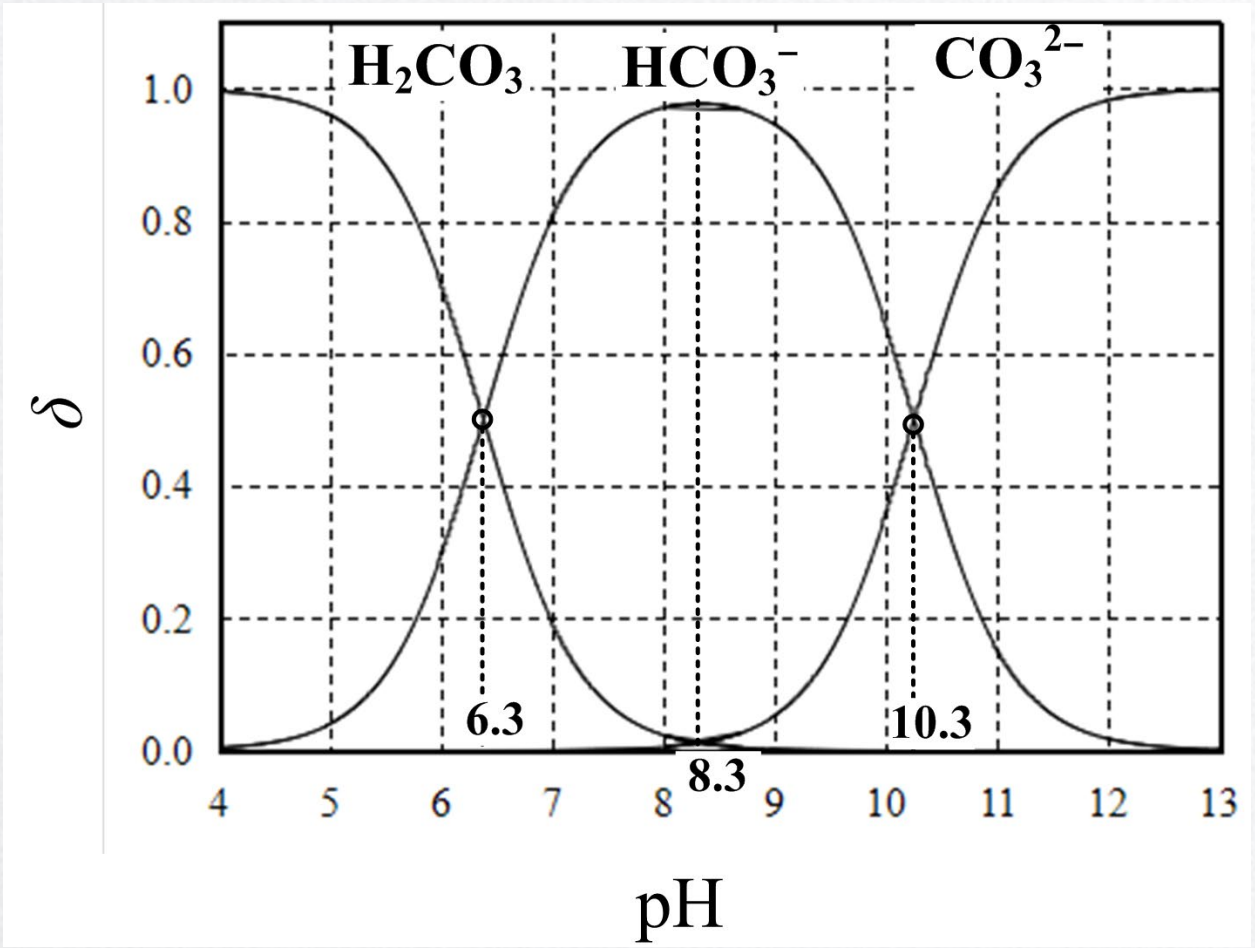
3.4 Calculation of pH – amphiprotic solution containing HA^-

$$H_2CO_3 \overset{+H^+}{\Longleftrightarrow} HCO_3^- \overset{-H^+}{\Longleftrightarrow} CO_3^{2-}$$

$$H_2PO_4^- \overset{+H^+}{\Longleftrightarrow} HPO_4^{2-} \overset{-H^+}{\Longleftrightarrow} PO_4^{3-}$$

$$H_3PO_4 \overset{+H^+}{\Longleftrightarrow} H_2PO_4^- \overset{-H^+}{\Longleftrightarrow} HPO_4^{2-}$$

3.4 Calculation of pH – amphiprotic solution containing HA^-



$$H_2CO_3 \overset{K_{a_1}}{\Longleftrightarrow} HCO_3^- \overset{K_{a_2}}{\Longleftrightarrow} CO_3^{2-}$$

$$\downarrow$$

$$NaHCO_3$$

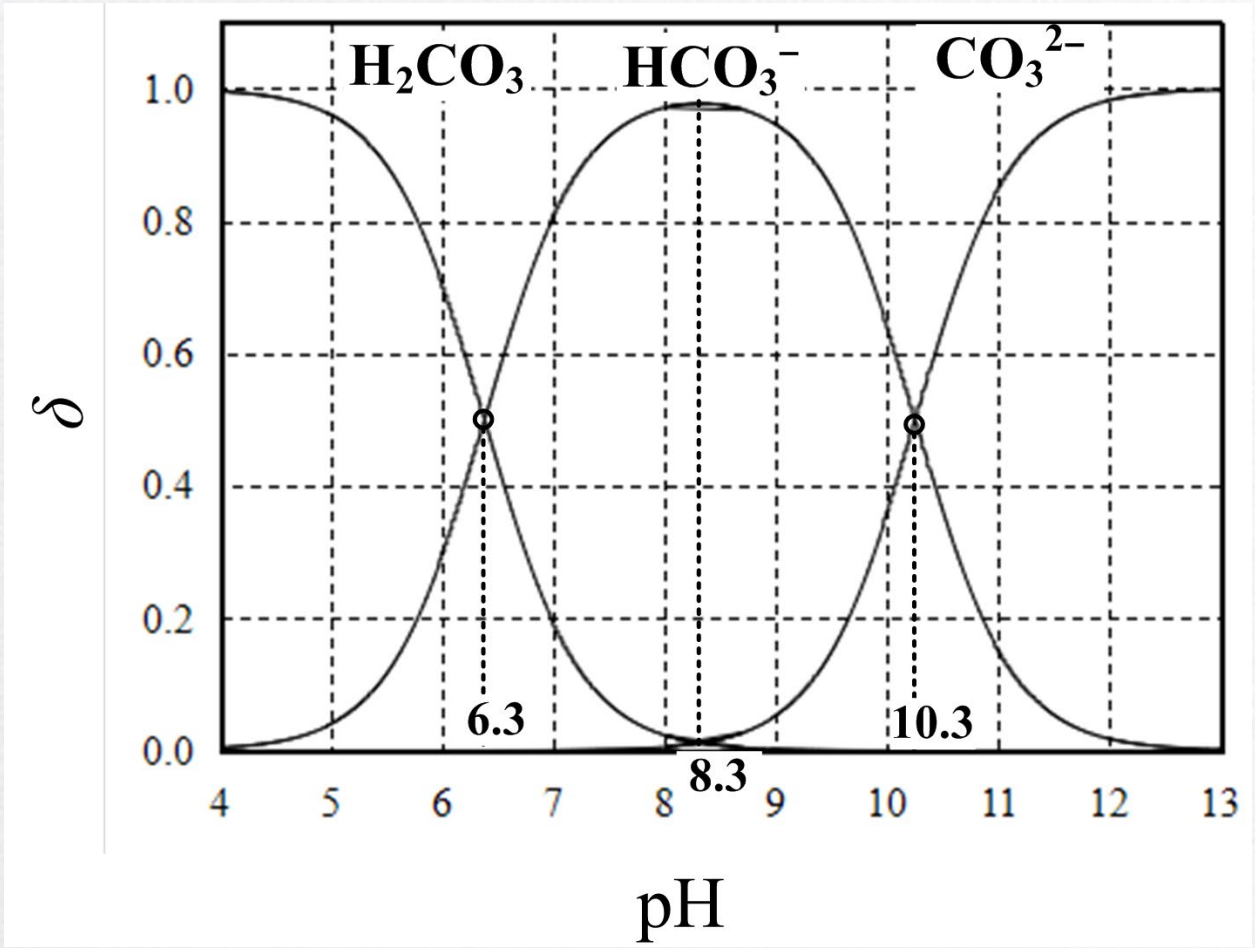
$$\downarrow$$

$$[H^+] \approx \sqrt{K_{a_1}K_{a_2}}$$

$$pH \approx \frac{pK_{a_1} + pK_{a_2}}{2}$$

$$pH \approx \frac{6.38 + 10.25}{2} = 8.31$$

3.4 Calculation of pH – amphiprotic solution containing HA^-



$$H_2CO_3 \overset{K_{a_1}}{\Longleftrightarrow} HCO_3^- \overset{K_{a_2}}{\Longleftrightarrow} CO_3^{2-}$$

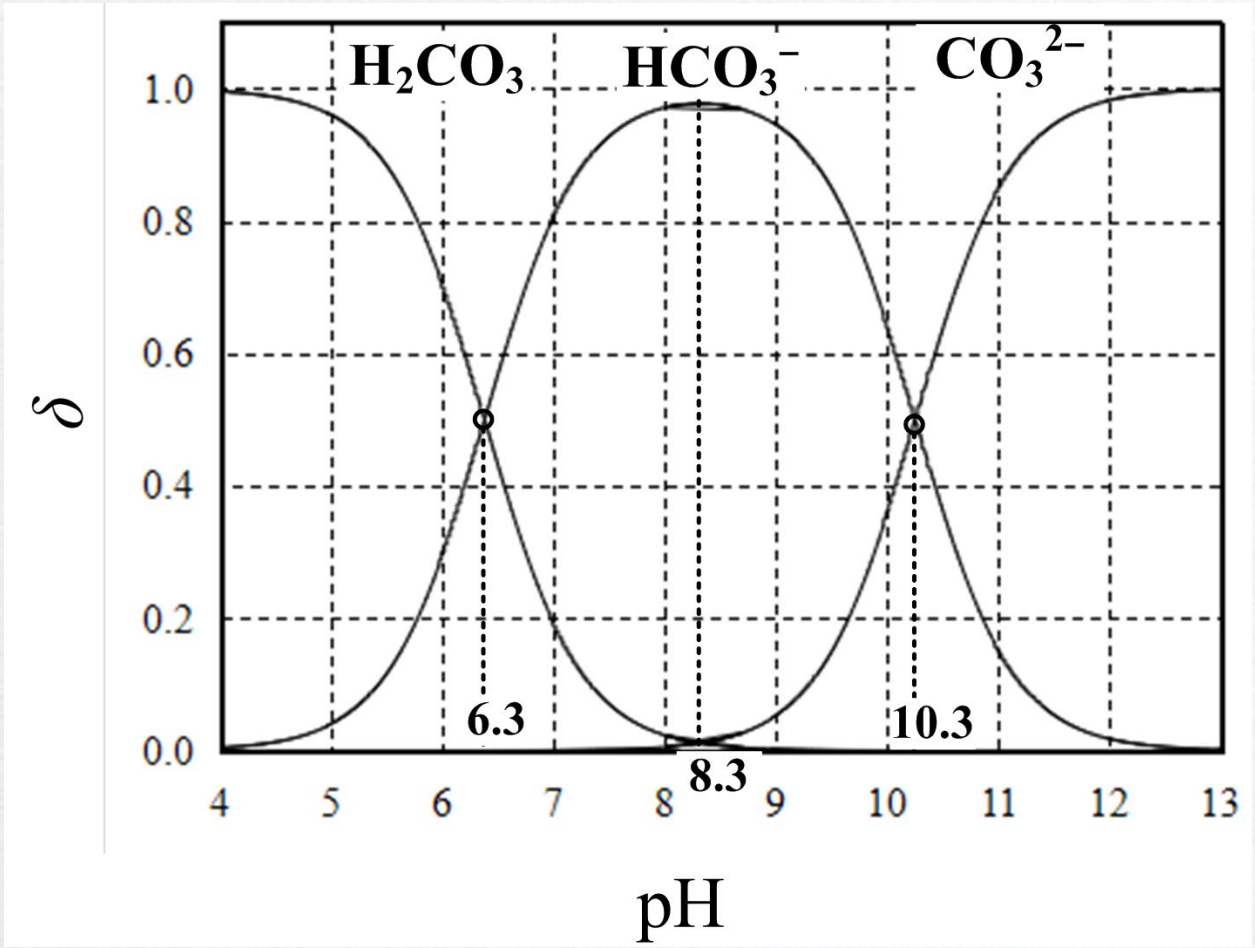
Na_2CO_3

$$K_{b_1} = \frac{K_w}{K_{a_2}} = 1.8 \times 10^{-4}$$

$$K_{b_2} = \frac{K_w}{K_{a_1}} = 2.3 \times 10^{-8}$$

$$\frac{K_{b_1}}{K_{b_2}} = 7.8 \times 10^3 \geq 100$$

3.4 Calculation of pH – amphiprotic solution containing HA^-



$$H_2CO_3 \overset{K_{a_1}}{\Longleftrightarrow} HCO_3^- \overset{K_{a_2}}{\Longleftrightarrow} CO_3^{2-}$$

Na_2CO_3

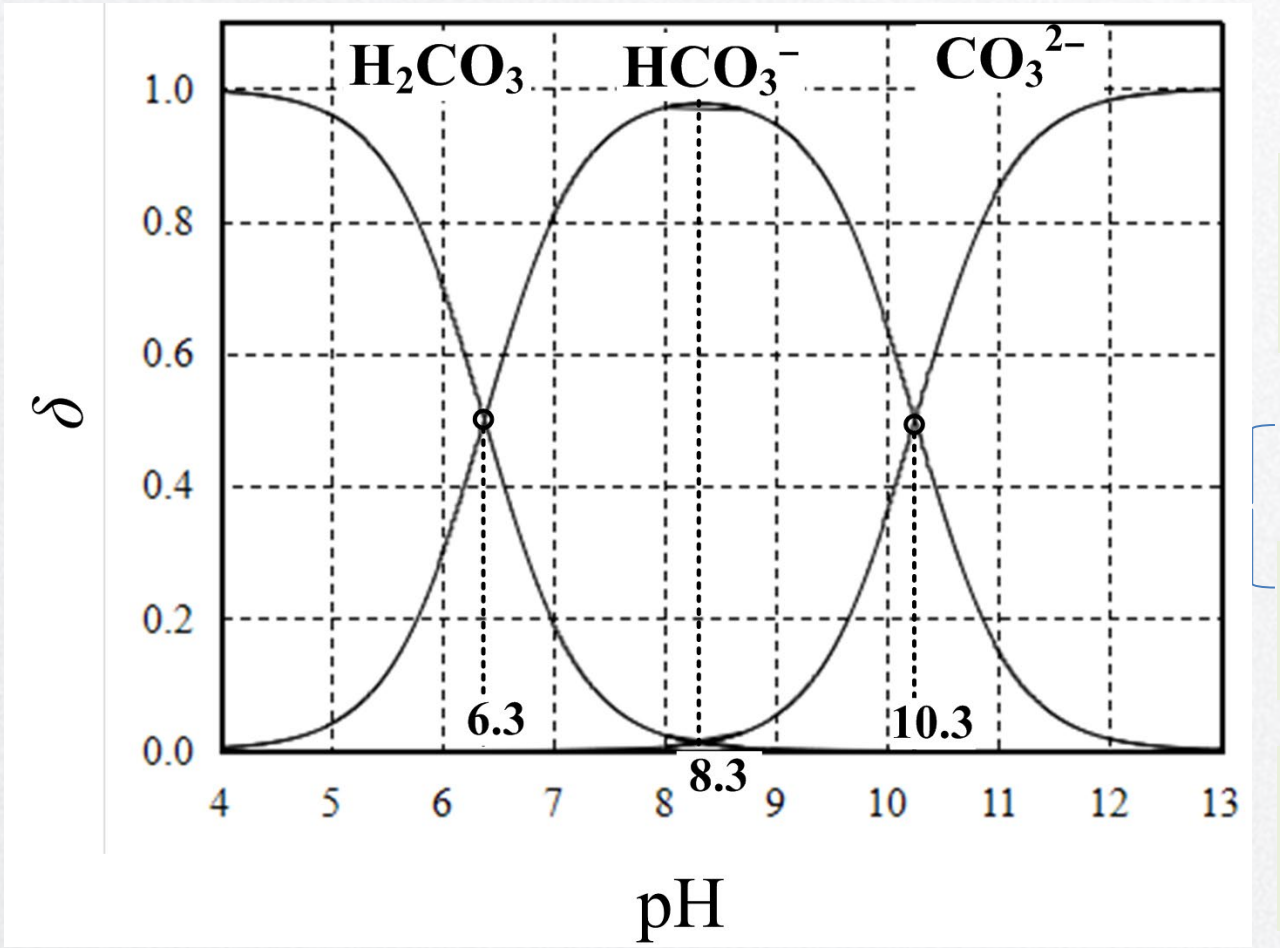
$$c_b K_{b1} = 0.10 \times 1.8 \times 10^{-4} \geq 20 K_w$$

$$\frac{c_b}{K_{b1}} = \frac{0.10}{1.8 \times 10^{-4}} \geq 500$$

$$[OH^-] = \sqrt{K_{b_1} \times c_b} = 0.42 \times 10^{-3}$$

$$pH = 14 - pOH = 14 - 3.38 = 10.62$$

3.4 Calculation of pH – amphiprotic solution containing HA^-



$$H_2CO_3 \overset{K_{a_1}}{\Longleftrightarrow} HCO_3^- \overset{K_{a_2}}{\Longleftrightarrow} CO_3^{2-}$$

$$\frac{K_{a_1}}{K_{a_2}} = \frac{4.2\times10^{-7}}{5.6\times10^{-11}} = 7.5\times10^{3} \geq 100$$

$$c_a K_{a1} = 0.04\times4.2\times10^{-7} \geq 20K_w$$

$$\frac{c_a}{K_{a1}} = \frac{0.04}{4.2\times10^{-7}} \geq 500$$

$$[H^+] = \sqrt{K_{a_1}\times c_a} = 1.30\times10^{-4}$$

$$pH = -\lg\left(1.30\times10^{-4}\right) = 3.89$$

3.4 Calculation of pH – Buffer solution

- **A buffered solution resists changes in pH when acid or bases are added or when dilution occurs.**
  - **Henderson-Hasselbalch equation**

$$pH = pK_a + lg\frac{[A^-]}{[HA]}$$

$$pH = pK_a + lg\frac{[B]}{[BH^+]}$$

$$pH = 4.76 + lg\frac{[Ac^-]}{[HAc]}$$

$$pH = 9.25 + lg\frac{[NH_3]}{[NH_4^+]}$$

Using this equation, it is possible to determine the degree of ionization of an acid at a given pH.

3.4 Calculation of pH – Buffer solution

- **Find the pH of a solution prepared by dissolving 12.43 g of tris- (MW 121.136) plus 4.67 g of tris-hydrochloride (MW 157.597) in 1.00 L water. (pK_a = 8.075)**

$$[B] = \frac{12.43}{121.136} = 0.1026\ mol/L \qquad [BH^+] = \frac{4.67}{157.597} = 0.0296\ mol/L$$

$$pH = pK_a + lg\frac{[B]}{[BH^+]} = 8.075 + lg\frac{0.1026}{0.0296} = 8.61$$

3.4 Calculation of pH – Buffer solution

- Calculate the pH of a buffer solution containing 1.0 mol/L HAc and 1.0 mol/L NaAc. (pKa = 4.76)
- Estimate the pH change when 1 ml of 6 mol/L HCl solution is added into 100 ml of this buffer solution.

Ac⁻

HAc

$$pH = pK_a + log\frac{[Ac^-]}{[HAc]} \qquad pH = 4.76 + \log\frac{1.0}{1.0} = 4.76$$

Initial $n_{HAc} = cv = \frac{1.0mmol}{ml} \times 100ml = 100mmol$

When HCl is added $n'_{HAc} = initial + produced = 100 + 6 = 106mmol$

Initial $n_{Ac^-} = cv = \frac{1.0mmol}{ml} \times 100ml = 100mmol$

When HCl is added $n'_{Ac^-} = initial - consummed = 100 - 6 = 94mmol$

$$pH = pK_a + log\frac{[Ac^-]}{[HAc]} = 4.76 + log\frac{94}{106} = 4.71$$

When 1 ml of 6 mol/L HCl solution is added.

$$HCl + Ac^- \rightarrow HAc + Cl^-$$

$$\downarrow \qquad \uparrow$$

6 mmol 6 mmol

v_{HCl} = 1 ml

c_{HCl} = 6 mol/l

$n_{HCl} = c \times v$

= 6 mmol

Homework

- Estimate the pH change when 1 ml of 6 mol/L NaOH solution is added into 100 ml of this buffer solution.

4 Titration Curve

Whether the analyte can be titrated accurately or not.

- pH change during the titration
  - Shape of titration curve
  - End point break
  - Choose correct indicator
    - Titration of strong acids and bases
    - Titration of monoprotic weak acids and bases
    - Titration of polyprotic acids and bases

4.1 Strong acid – base titration

We use **titration constant**（K_t）to evaluate the degree of the reaction.

$$H^+ + OH^- \leftrightarrow H_2O$$

$$K_t = \frac{1}{[H^+][OH^-]} = \frac{1}{K_w} = 1.0 \times 10^{14}$$

4.1 Strong acid – base titration

0.10 mol/L NaOH

- **A strong acid** is titrated **by a strong base**.
- **NaOH（0.10mol/L）→ HCl（ 0.10 mol/L, 20 mL ）**
  - – ***pH change*** during the titration
  - – The ***shape*** of the titration curve
  - – The ***end point break***
  - – The ***impact factors*** of the end point break
  - – Choose correct ***indicator***



20 ml
0.10 mol/L HCl

4.1 Strong acid – base titration

pH change during the titration

- Analyte： HCl, V_{HCl} = 20 ml, c_{HCl} = 0.10 mol/L
- Titrant： NaOH, V_{NaOH} = *x* mL, c_{NaOH} = 0.10 mol/L

$$NaOH + HCl \rightarrow NaCl + H_2O$$

- At the beginning of the titration
  - 20 mL, 0.10 mol/L, HCl

$$pH = -\lg[H^+] = -\lg c_{HCl} = -\lg 0.10 = 1.00$$



4.1 Strong acid – base titration

pH change during the titration

- **At the beginning of the titration** $pH = 1.00$
- **Before reaching the equivalence point** $NaOH + HCl \rightarrow NaCl + H_2O$
  - **HCl + NaCl** in the solution and $v_{HCl} > v_{NaOH}$
  - pH value depends on the **amount of HCl left** and the total volume of the solution.

$$[H^+] = \frac{n_{HCl} - n_{NaOH}}{V_{HCl} + V_{NaOH}} = \frac{c_{HCl}V_{HCl} - c_{NaOH}V_{NaOH}}{V_{HCl} + V_{NaOH}}$$



4.1 Strong acid – base titration

pH change during the titration

- **At the beginning of the titration** $pH = 1.00$
- **Before reaching the equivalence point** $NaOH + HCl \rightarrow NaCl + H_2O$
  - When **19.98 ml** of NaOH is added.

$$\frac{19.98ml}{20.00ml} \times 100\% = 99.9\%,$$

It means **0.1% before reaching the equivalence point.**



4.1 Strong acid – base titration

pH change during the titration

- **At the beginning of the titration** $pH = 1.00$
- **Before reaching the equivalence point** $NaOH + HCl \rightarrow NaCl + H_2O$
  - When **19.98 ml** of NaOH is added.

$$[H^+] = \frac{0.10 \times 20.00 - 0.10 \times 19.98}{20.00 + 19.98} = 5.00 \times 10^{-5}$$

$$pH = -\lg[H^+] = -\lg(5.00 \times 10^{-5}) = 4.30$$



4.1 Strong acid – base titration

pH change during the titration

- At the beginning of the titration $pH = 1.00$
- Before reaching the equivalence point $pH = 4.30$
- At the equivalence point $NaOH + HCl \rightarrow NaCl + H_2O$
  - Only NaCl in the solution and $V_a = V_b$
  - $[H^+] = [OH^-] = 1.00 \times 10^{-7}$ mol/L

$$pH = -\lg[H^+] = -\lg 10^{-7} = 7.00$$



0.10 mol/L NaOH

20 ml
0.10 mol/L HCl

4.1 Strong acid – base titration

pH change during the titration

- At the beginning of the titration $pH = 1.00$
- Before reaching the equivalence point $pH = 4.30$
- At the equivalence point $pH = 7.00$
- After the equivalence point $NaOH + HCl \rightarrow NaCl + H_2O$
  - NaOH + NaCl in the solution and $V_{NaOH} > V_{HCl}$
  - The pH of the solution depends on the amount of the NaOH and the volume of the solution.
  - 20.02 mL NaOH was added, 0.1% after the equivalence point.



4.1 Strong acid – base titration

pH change during the titration

- At the beginning of the titration $pH = 1.00$
- Before reaching the equivalence point $pH = 4.30$
- At the equivalence point $pH = 7.00$
- After the equivalence point $NaOH + HCl \rightarrow NaCl + H_2O$

$$[OH^-] = \frac{0.10 \times 20.02 - 0.10 \times 20.00}{20.02 + 20.00} = 5.00 \times 10^{-5}$$

$$pH = -\lg[H^+] = -\lg\frac{K_w}{[OH^-]} = 9.70$$



4.1 Strong acid – base titration

pH change during the titration

		V_{NaOH}	
• At the beginning of the titration	$pH = 1.00$	0 ml	
• Before reaching the equivalence point	$pH = 4.30$	19.98 ml	- 0.1%
• At the equivalence point	$pH = 7.00$	20.00 ml	
• After the equivalence point	$pH = 9.70$	20.02 ml	+ 0.1%

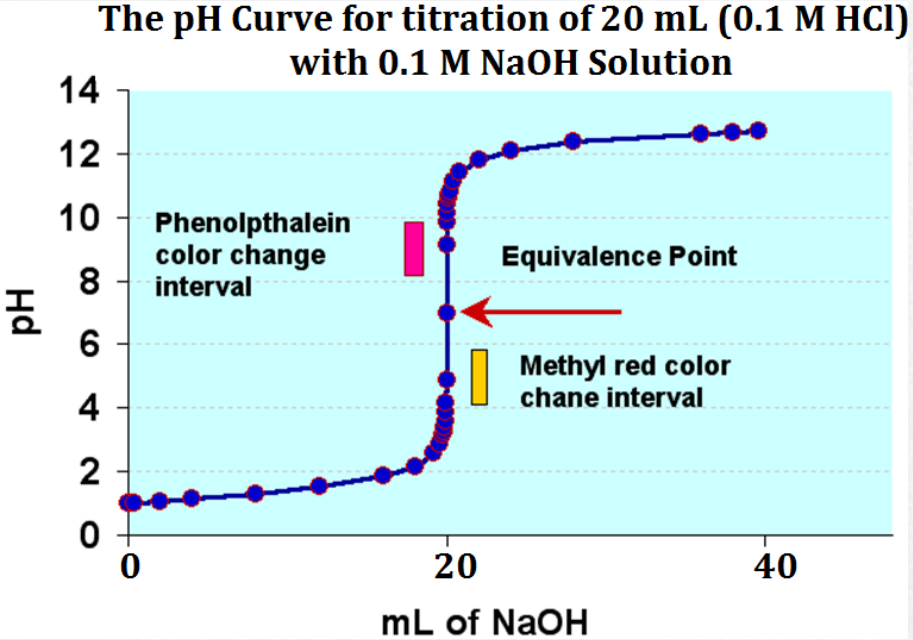
Shape of titration curve

- **Titration curve**
  - The curve to describe the ***relationship*** between the **titrant volume** and the **solution parameter** during the process of the titration.

4.1 Strong acid – base titration

Shape of titration curve

End point break



The sudden change of pH near the equivalence point.

Range of end point break: ± 0.1% of equivalence point.

Thus, it is from pH 4.30 to pH 9.70

4.1 Strong acid – base titration

The impact factors of the end point break

c_a = 1.0 mol/L, **ΔpH** = 3.30 - 10.70

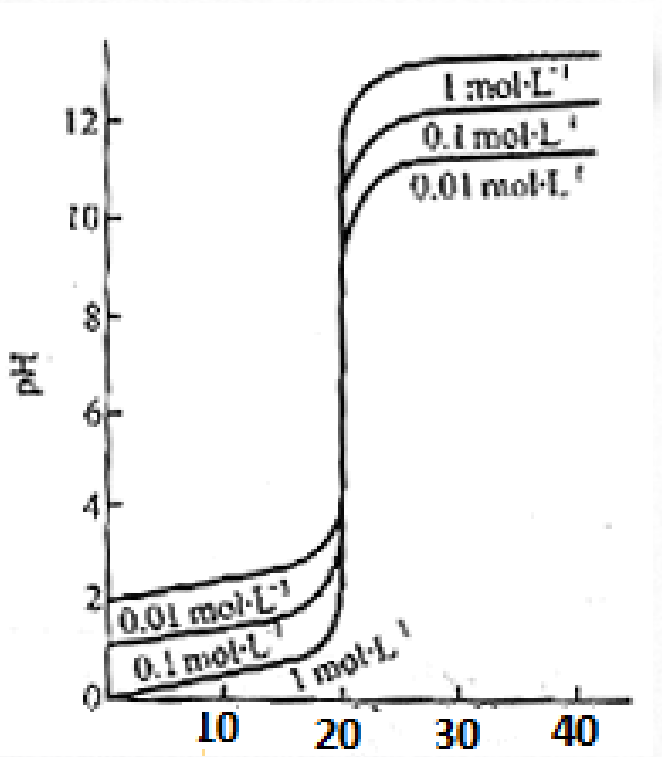
- Methyl orange, methyl red and phenolphthalein

c_a = 0.10 mol/L, **ΔpH** = 4.30 - 9.70

- Methyl orange (**poor**), methyl red and phenolphthalein

c_a = 0.010 mol/L, **ΔpH** = 5.30 - 8.70

- methyl red and phenolphthalein

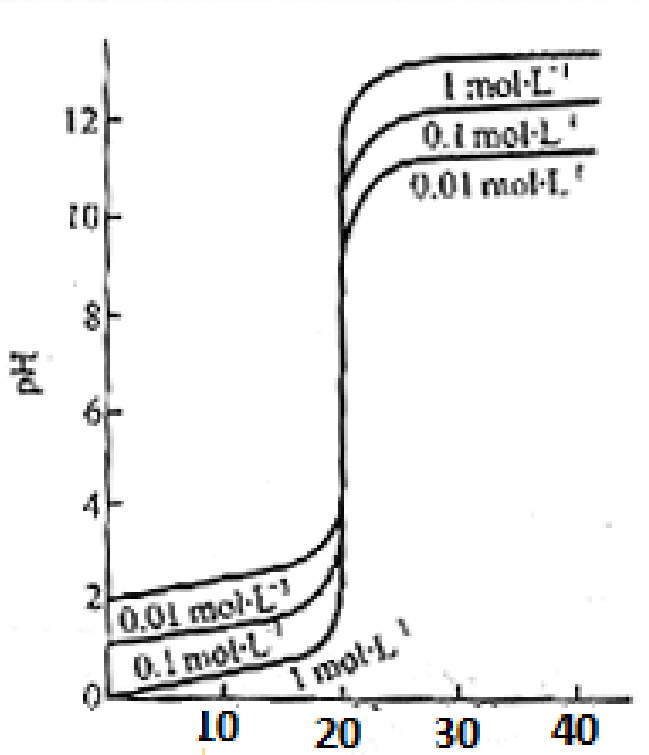


The indicators

Methyl orange（3.1~4.4）**3.45
Methyl red（4.4~6.2）**5.1
Phenolphthalein（8.0~10.0）**9.1

4.1 Strong acid – base titration

The impact factors of the end point break



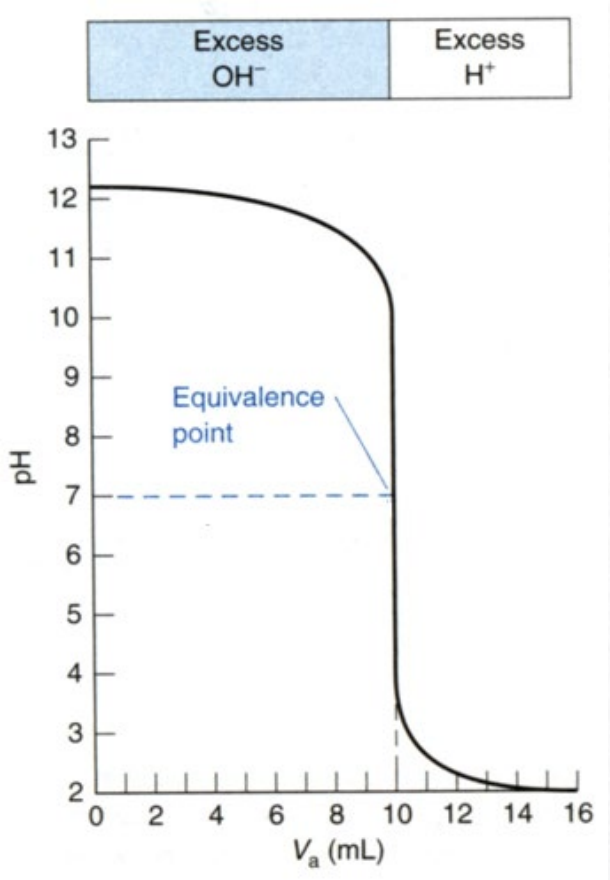
Impact factor:

Concentration of the acid and base

- $c \uparrow, \Delta pH \uparrow$, more choices of indicators
- Example, when $c \uparrow 10$ times, $\Delta pH \uparrow 2$ pH units.

4.1 Strong acid – base titration

- HCl（0.10mol/L）→ NaOH（ 0.10 mol/L, 20 mL ）
  - Similar titration curve obtained and pH change is reversed
  - The end point break depends on the concentration of the standard solution
  - Appropriate indicator



0.10 mol/L HCl



20 ml
0.10 mol/L NaOH

4.2 Weak acid – base titration

0.10 mol/L NaOH

- **A weak acid is titrated by a strong base.**
- **NaOH（0.10mol/L）→ HAc（ 0.10 mol/L, 20 mL ）**
  - ***pH change*** during the titration
  - The ***shape*** of the titration curve
  - The ***end point break***
  - The ***impact factors*** of the end point break
  - Choose correct ***indicator***



Weak acid

20 ml

0.10 mol/L HAc

4.2 Weak acid – base titration

0.10 mol/L NaOH

pH change during the titration

$$NaOH + HAc \rightarrow NaAc + H_2O$$

- **Analyte: HAc**, V_{HCl} = 20 ml, c_{HCl} = 0.10 mol/L
- **Titrant: NaOH**, V_{NaOH} = *x* mL, c_{NaOH} = 0.10 mol/L
- **At the beginning of the titration**

Weak acid

- **20 mL, 0.10 mol/L, HAc**

$$c_a K_a = 0.10 \times 10^{-4.76} > 20K_w \qquad \frac{c_a}{K_a} = \frac{0.10}{10^{-4.76}} = 10^{3.76} > 500$$

$$[H^+] = \sqrt{c_a K_a} = \sqrt{0.10 \times 10^{-4.76}} = 0.00132$$

$$pH = -\lg[H^+] = -\lg 0.00132 = 2.88$$



20 ml
0.10 mol/L HAc

4.2 Weak acid – base titration

0.10 mol/L NaOH

pH change during the titration

$$NaOH + HAc \rightarrow NaAc + H_2O$$

- At the beginning of the titration

$$pH = 2.88$$

- Before reaching the equivalence point
  - HAc + NaAc in the solution and $V_{HAc} > V_{NaOH}$
  - pH value depends on HAc ~ NaAc ***buffer solution***

$$pH = pK_a + lg\frac{[Ac^-]}{[HAc]}$$



20 ml
0.10 mol/L HAc

4.2 Weak acid – base titration

0.10 mol/L NaOH



20 ml
0.10 mol/L HAc

pH change during the titration

- At the beginning of the titration $pH = 2.88$
- Before reaching the equivalence point
  - When 19.98 ml of 0.10 mol/L NaOH is added.

$$NaOH + HAc \rightarrow NaAc + H_2O$$

$$pH = pK_a + lg\frac{[Ac^-]}{[HAc]} = pK_a + lg\frac{n_{NaAc}}{n_{HAc}^{left}}$$

$$pH = 4.76 + lg\frac{0.10 \times 19.98}{0.10 \times 20.00 - 0.10 \times 19.98} = 7.76$$

4.2 Weak acid – base titration

0.10 mol/L NaOH

pH change during the titration

- At the beginning of the titration $pH = 2.88$
- Before reaching the equivalence point $pH = 7.76$
- At the equivalence point
  - Only NaAc in the solution and $V_a = V_b$

$$NaOH + HAc \rightarrow NaAc + H_2O$$

  - pH of the solution depends on the dissociation of Ac^-.



20 ml
0.10 mol/L HAc

4.2 Weak acid – base titration

0.10 mol/L NaOH

pH change during the titration

- At the beginning of the titration $pH = 2.88$
- Before reaching the equivalence point $pH = 7.76$
- At the equivalence point
  - As the volume has been doubled

$$c_{Ac^-} = \frac{c_{NaOH}}{2} = \frac{0.10}{2} = \mathbf{0.050 mol/L}$$



20 ml
0.10 mol/L HAc

4.2 Weak acid – base titration

0.10 mol/L NaOH

pH change during the titration

- At the beginning of the titration $pH = 2.88$
- Before reaching the equivalence point $pH = 7.76$
- At the equivalence point

$$K_b = \frac{K_w}{K_a} = \frac{10^{-14}}{10^{-4.76}} = 10^{-9.24}$$

$$c_b K_b = 0.050 \times 10^{-9.24} > 20K_w \qquad \frac{c_b}{K_b} = \frac{0.050}{10^{-9.24}} > 500$$



20 ml
0.10 mol/L HAc

4.2 Weak acid – base titration

0.10 mol/L NaOH

pH change during the titration

- At the beginning of the titration $pH = 2.88$
- Before reaching the equivalence point $pH = 7.76$
- At the equivalence point

$$[OH^-] = \sqrt{c_b K_b} = \sqrt{0.050 \times 10^{-9.24}} = 5.37 \times 10^{-6}$$

$$pOH = -\lg[OH^-] = 5.27$$

$$pH = 14 - 5.27 = 8.73$$



20 ml
0.10 mol/L HAc

4.2 Weak acid – base titration

0.10 mol/L NaOH



20 ml
0.10 mol/L HAc

pH change during the titration

- At the beginning of the titration $pH = 2.88$
- Before reaching the equivalence point $pH = 7.76$
- At the equivalence point $pH = 8.73$
- After the equivalence point
  - NaOH + NaAc in the solution and $V_{NaOH} > V_{HAc}$
  - The pH of the solution depends on concentration of the NaOH and dissociation of NaAc.

Suppressed by strong base NaOH

4.2 Weak acid – base titration

0.10 mol/L NaOH

pH change during the titration

- At the beginning of the titration $pH = 2.88$
- Before reaching the equivalence point $pH = 7.76$
- At the equivalence point $pH = 8.73$
- After the equivalence point
  - 20.02 mL NaOH was added, 0.1% after the equivalence point.

$$[OH^-] = \frac{0.10 \times 20.02 - 0.10 \times 20.00}{20.02 + 20.00} = 5.00 \times 10^{-5}$$

$$pH = -\lg[H^+] = -\lg\frac{K_W}{[OH^-]} = 9.70$$



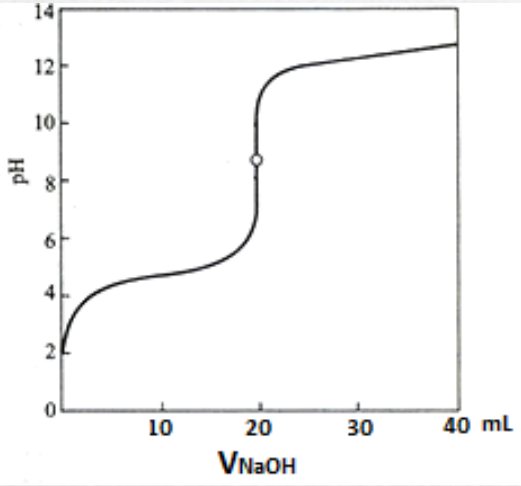
20 ml
0.10 mol/L HAc

4.2 Weak acid – base titration

pH change during the titration

V_{NaOH}

- At the beginning of the titration — $pH = 2.88$ — 0 ml
- Before reaching the equivalence point — $pH = 7.76$ — 19.98 ml — - 0.1%
- At the equivalence point — $pH = 8.73$ — 20.00 ml
- After the equivalence point — $pH = 9.70$ — 20.02 ml — + 0.1%



pH changes as NaOH is added to HAc

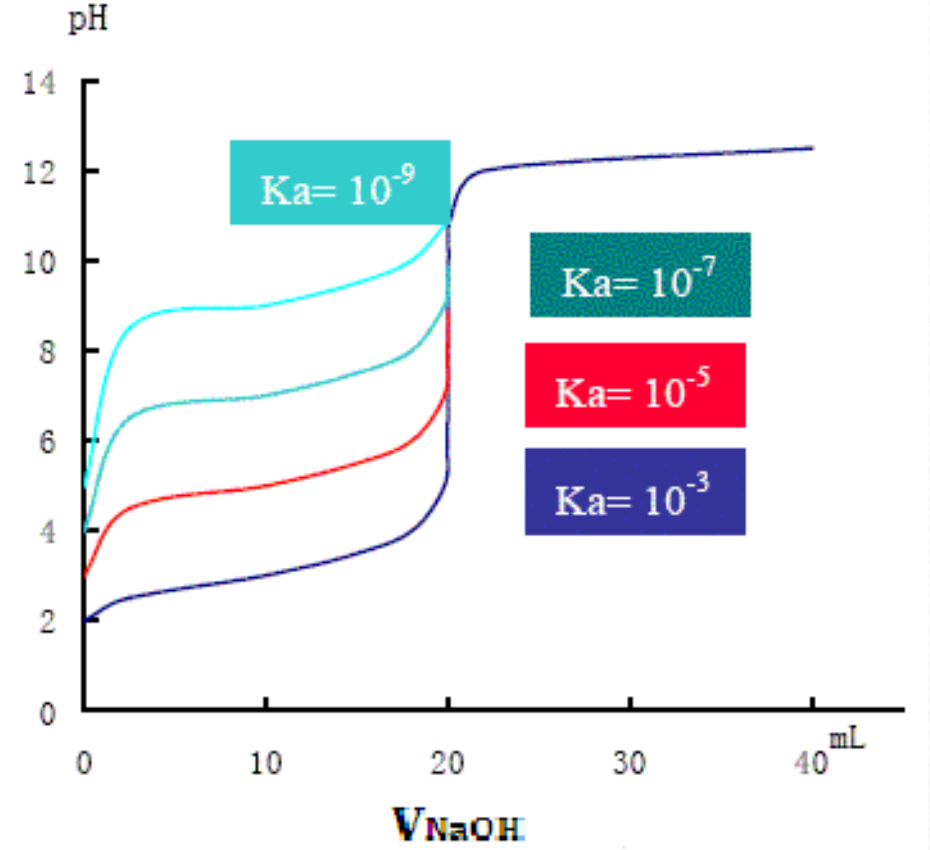
The indicators

- Methyl orange（3.1~4.4）**3.45 ×
- Methyl red（4.4~6.2）**5.1 ×
- Phenolphthalein（8.0~10.0）**9.1

4.2 Weak acid – base titration

The impact factor of the rang of end point break

- K_a – Acidity (c_a is fixed)
  - *Higher* value of K_a with stronger acidity has *larger* rang of end point break.
  - There will be no end point break when $K_a < 10^{-9}$. In this case, this acid can't be titrated by base in water, even when the c_a = 1.0 mol/L.



4.2 Weak acid – base titration

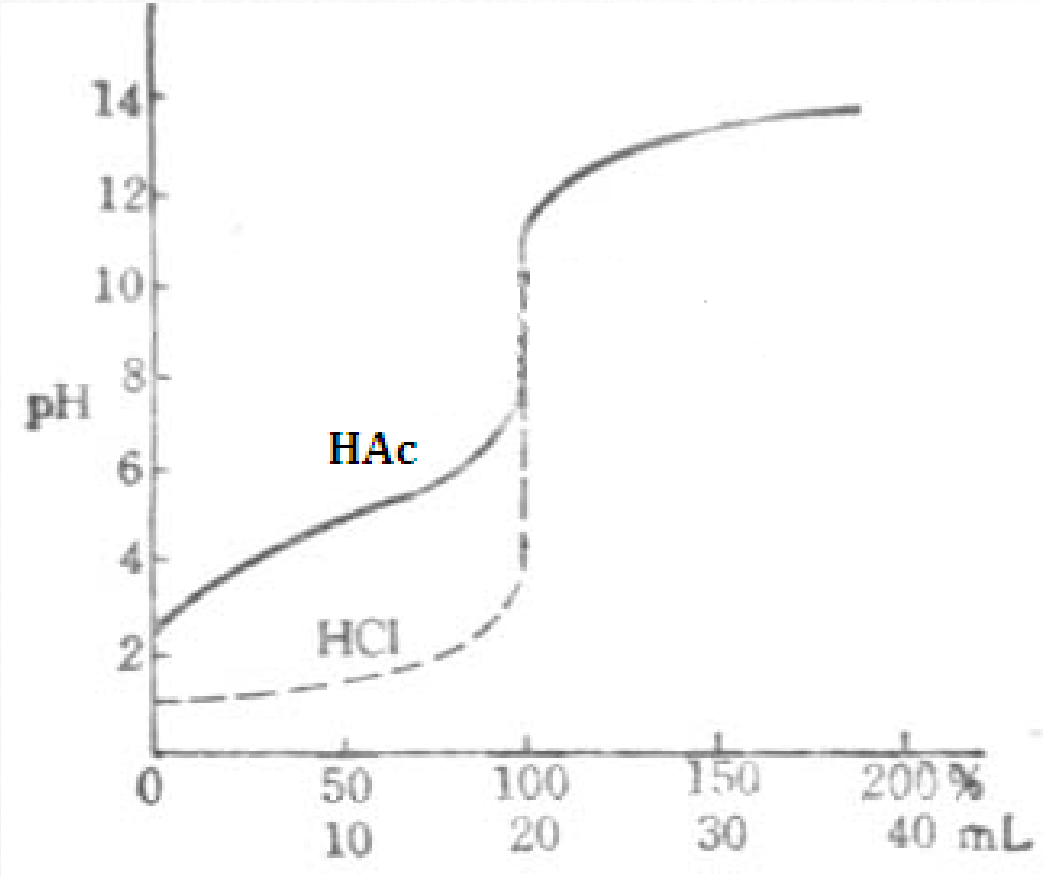
The impact factor of the rang of end point break

- c_a – Concentration of the acid (K_a is a constant)
  - The influence of the concentration to the end point break is as same as the strong acid-base titration.

- Normally, when $c_a K_a \geq 10^{-8}$, the acid can be titrated with base in water. (TE ≤ 0.1%)

4.2 Weak acid – base titration

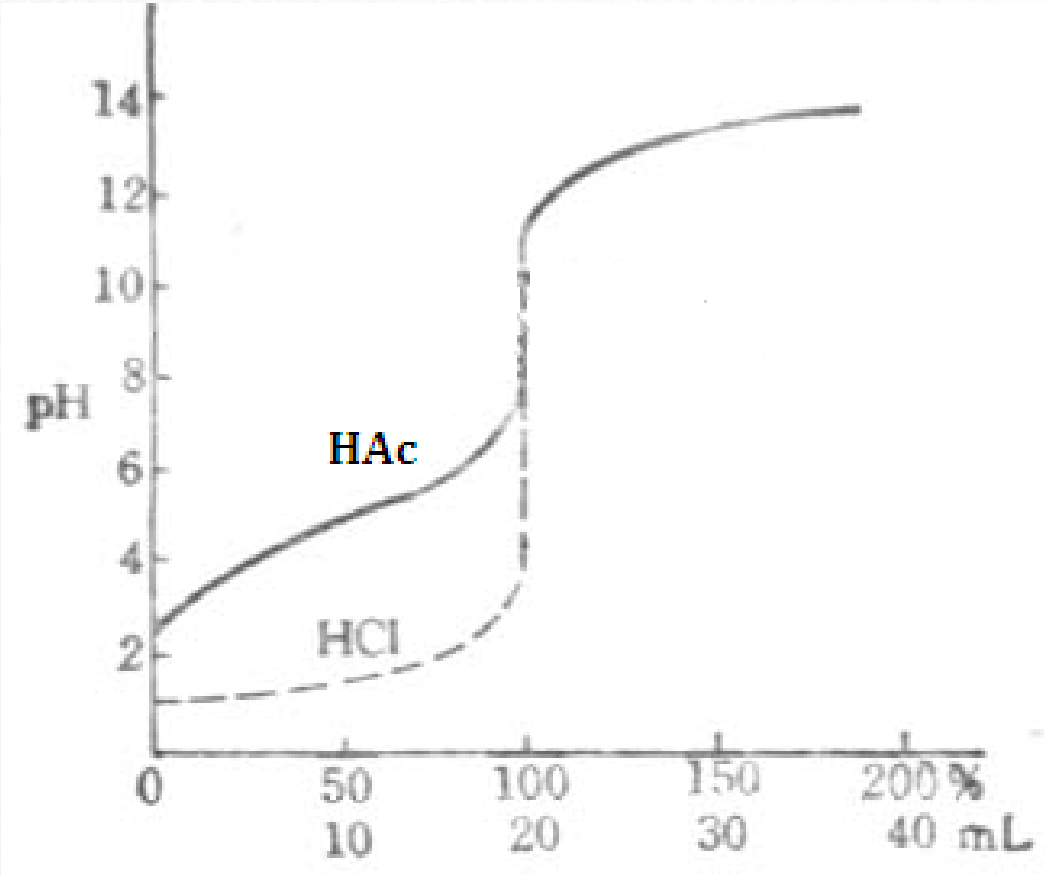
- HAc is a ***weak acid*** and partially dissociate in water.
- The ***starting point*** of the titration curve is **2.88**, a little ***higher*** than strong acid HCl for **2 pH units**.



4.2 Weak acid – base titration

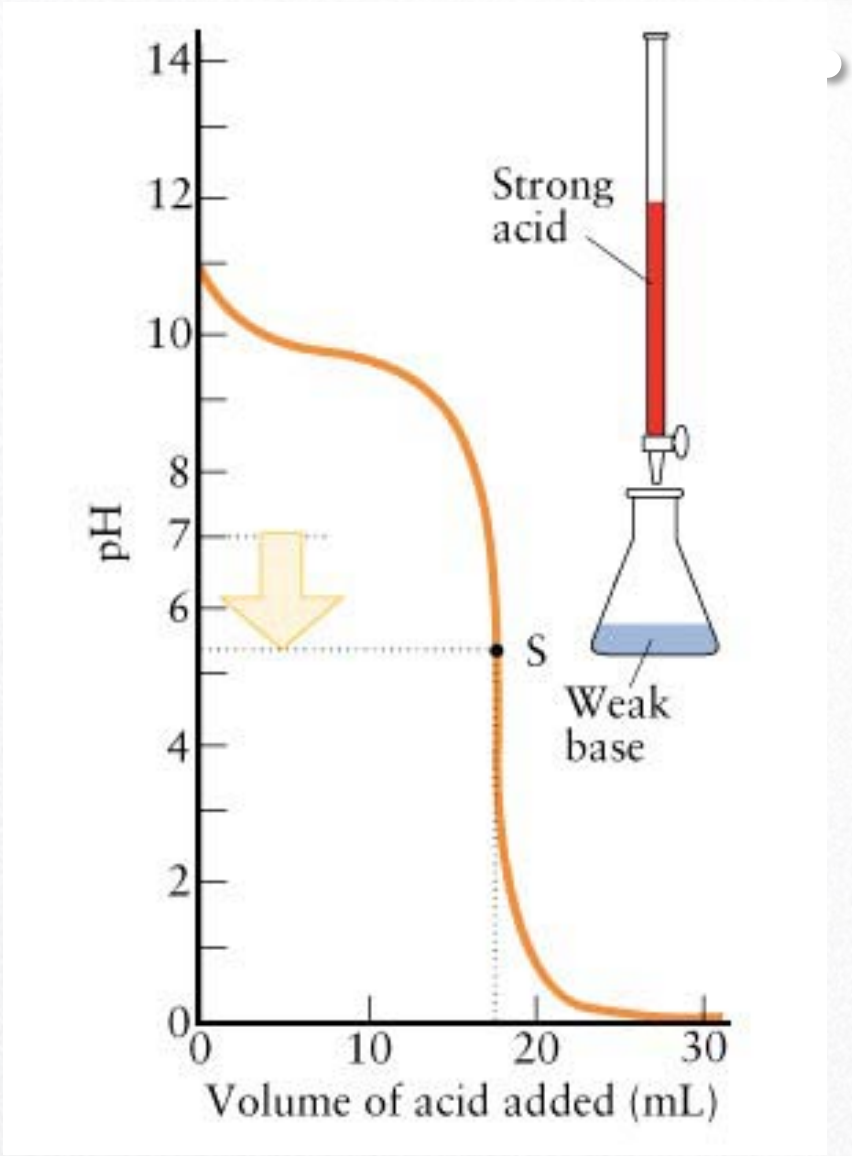
$$NaOH + HAc \rightarrow NaAc + H_2O$$

- At beginning of the titration, the inhibition of Ac^- to the dissociation of HAc, the pH changes fast, when ***buffer formed***, pH changes slow.
- When approaching to equivalence point, pH changes fast.
- The rang of end point break (7.76 to 9.70) is narrower than HCl/NaOH titration.



4.2 Weak acid – base titration

- Normally, when $c_b K_b \geq 10^{-8}$, the acid can be titrated with base in water. ($TE \leq 0.1\%$)

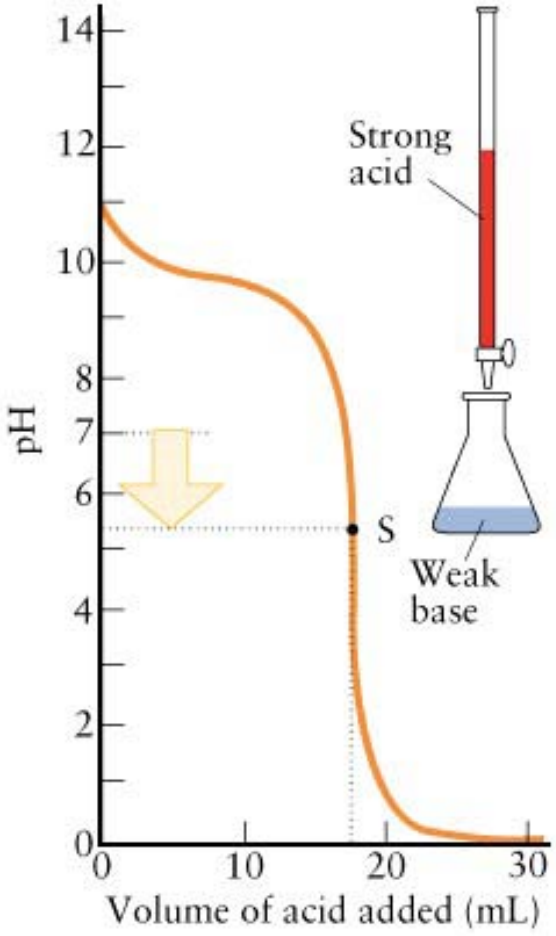


4.2 Weak acid – base titration

- HCl (0.10 mol/L) → $NH_3 \cdot H_2O$ (0.10 mol/L, 20.00 mL)。
  - $K_b = 1.76 \times 10^{-5}$
  - $c \cdot K_b = 1.76 \times 10^{-6} \geq 10^{-8}$

pH = 5.28

Δ pH = 6.43 ~ 4.30



- Methyl orange（3.1~4.4）**3.45
- Methyl red（4.4~6.2）**5.1
- Phenolphthalein（8.0~10.0）**9.1 ×

4.3 Polyprotic acids and bases titration

$$H_2A \overset{K_{a_1}}{\Longleftrightarrow} H^+ + HA^- \qquad HA^- \overset{K_{a_2}}{\Longleftrightarrow} H^+ + A^{2-}$$

If $c_a K_{a_1} \geq 10^{-8}$

The H^+ from the first step dissociation can be titrated directly.

If $\frac{K_{a_1}}{K_{a_2}} \geq 10^4$

The H^+ from the second step dissociation won't interfere the titration of the H^+ in first step.

The **first end point break can be observed**.

4.3 Polyprotic acids and bases titration

NaOH (0.10 mol/L) → H_3PO_4 (0.10 mol/L, 20.00 ml)

$$K_{a_1} = 10^{-2.16}$$

$$H_3PO_4 \overset{K_{a_1}}{\Longleftrightarrow} H^+ + H_2PO_4^-$$

$$c_a K_{a_1} \geq 10^{-8}$$

$$\frac{K_{a_1}}{K_{a_2}} \geq 10^4$$

Protons dissociated from the **first step can be titrated** accurately and the **first end point break can be observed**.

4.3 Polyprotic acids and bases titration

NaOH (0.10 mol/L) → H_3PO_4 (0.10 mol/L, 20.00 ml)

$$K_{a_2} = 10^{-7.21}$$

$$H_2PO_4^- \overset{K_{a_2}}{\Longleftrightarrow} H^+ + HPO_4^{2-}$$

$$c_a K_{a_2} \approx 10^{-8}$$

$$\frac{K_{a_2}}{K_{a_3}} \geq 10^4$$

Protons dissociated from the **second step can be titrated** accurately and **the second end point break can be observed**.

4.3 Polyprotic acids and bases titration

NaOH (0.10 mol/L) → H_3PO_4 (0.10 mol/L, 20.00 ml)

$K_{a_3} = 10^{-12.32}$

$$HPO_4^{2-} \overset{K_{a_3}}{\Longleftrightarrow} H^+ + PO_4^{3-}$$

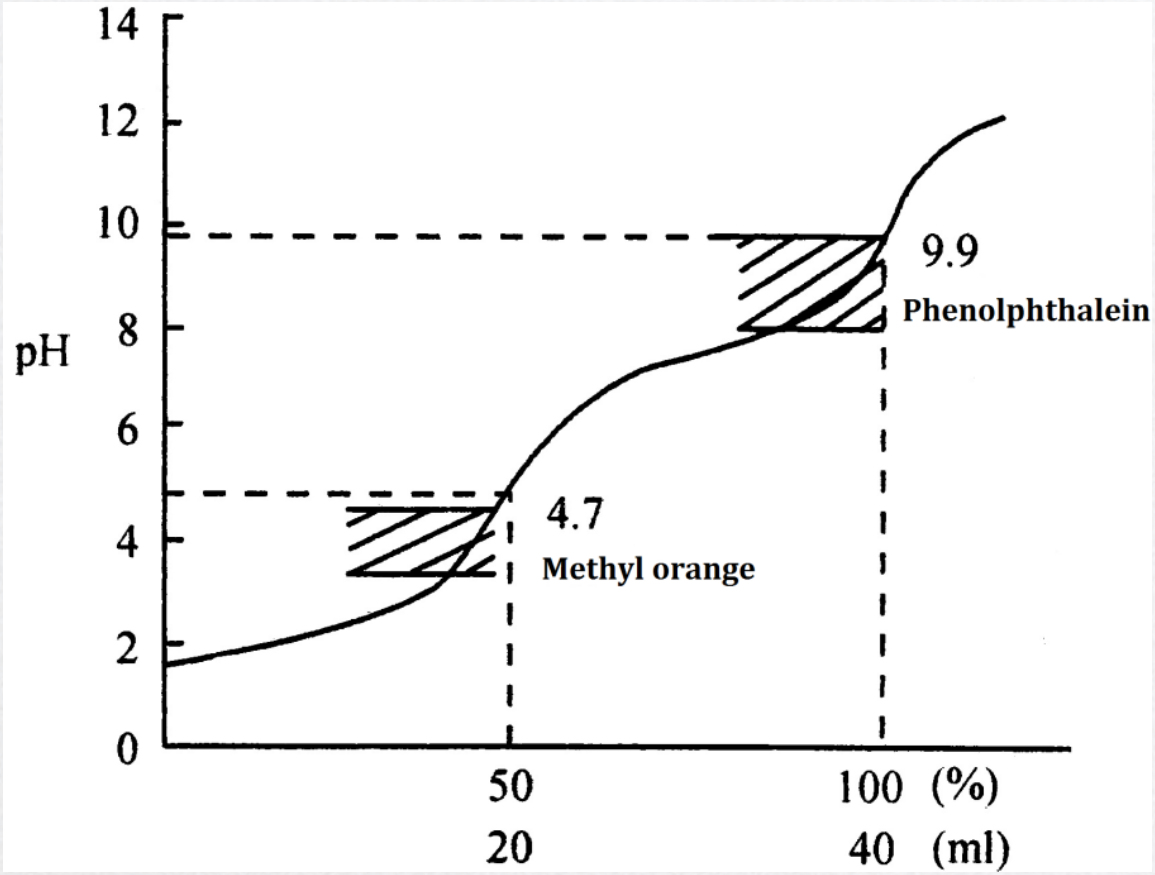
$c_a K_{a_3} < 10^{-8}$

Protons dissociated from the **third step can NOT be titrated** accurately and **NO end point break can be observed**.

4.3 Polyprotic acids and bases titration

$K_{a_1} = 10^{-2.16}$

$K_{a_2} = 10^{-7.21}$ $K_{a_3} = 10^{-12.32}$



$$NaOH + H_3PO_4 \rightarrow NaH_2PO_4 + H_2O$$

pH_{sp} depends on the dissociation of $H_2PO_4^-$.

$$[H^+] \approx \sqrt{K_{a_1} K_{a_2}}$$

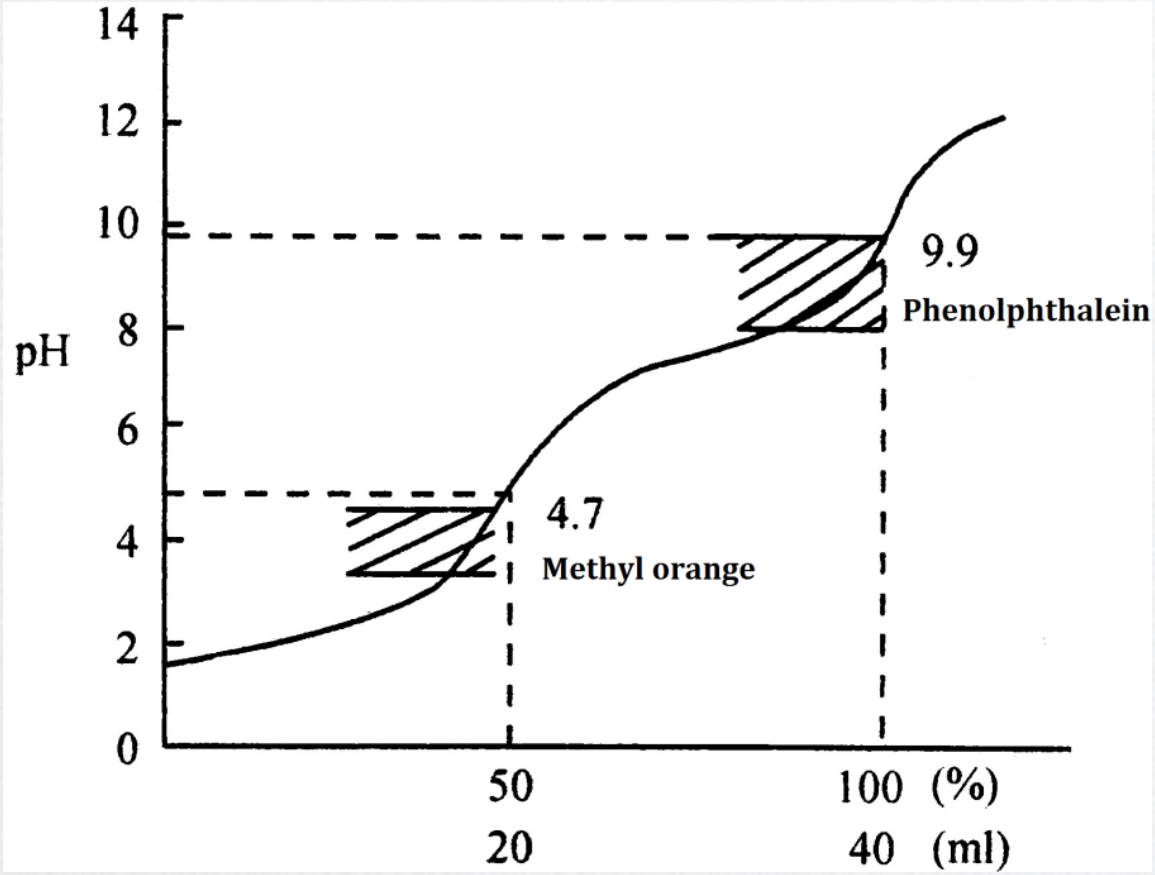
$$pH_{sp_1} \approx \frac{pK_{a_1} + pK_{a_2}}{2}$$

$$= \frac{2.16 + 7.21}{2} = 4.68$$

4.3 Polyprotic acids and bases titration

$K_{a_1} = 10^{-2.16}$

$K_{a_2} = 10^{-7.21}$ $K_{a_3} = 10^{-12.32}$



$$NaOH + H_3PO_4 \rightarrow NaH_2PO_4 + H_2O$$

$$NaOH + NaH_2PO_4 \rightarrow Na_2HPO_4 + H_2O$$

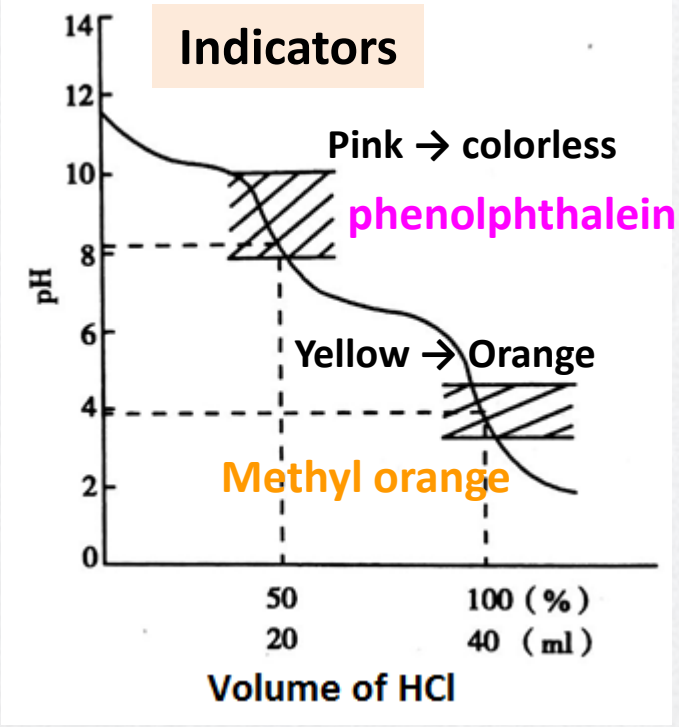
pH_{sp} depends on the dissociation of HPO_4^-.

$$pH_{sp_2} \approx \frac{pK_{a_2} + pK_{a_3}}{2}$$

$$= \frac{7.21 + 12.32}{2} = 9.76$$

4.3 Polyprotic acids and bases titration

HCl (0.10 mol/L) → Na_2CO_3 (0.10 mol/L, 20.00 ml)



$$K_{a_1} = 4.2 \times 10^{-7}$$

$$K_{b_1} = \frac{K_w}{K_{a_2}} = 1.8 \times 10^{-4}$$

$$c_b K_{b_1} \geq 10^{-8}$$

$$K_{a_2} = 5.6 \times 10^{-11}$$

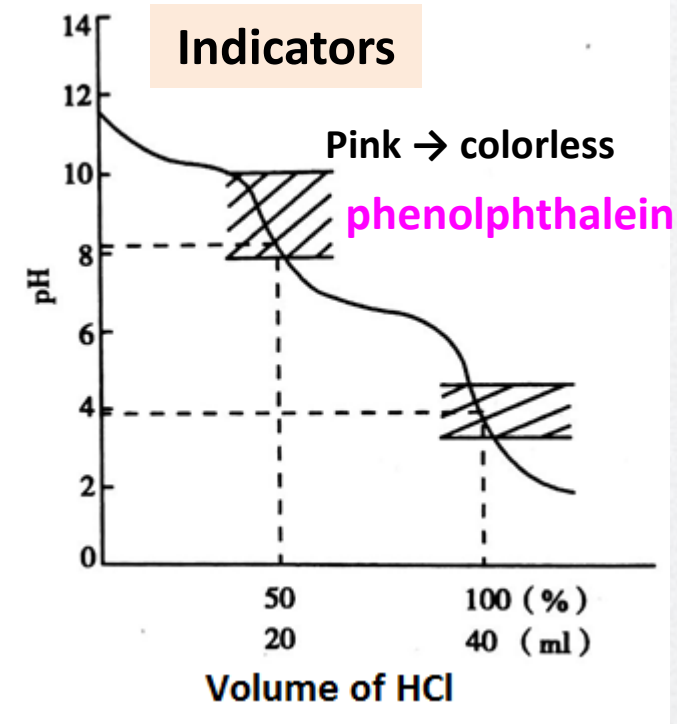
$$K_{b_2} = \frac{K_w}{K_{a_1}} = 2.3 \times 10^{-8}$$

$$\frac{K_{b_1}}{K_{b_2}} \approx 10^4$$

$$c_b K_{b_2} \geq 10^{-8}$$

4.3 Polyprotic acids and bases titration

HCl (0.10 mol/L) → Na_2CO_3 (0.10 mol/L, 20.00 ml)



Step 1: $HCl + Na_2CO_3 \rightarrow NaHCO_3 + NaCl$

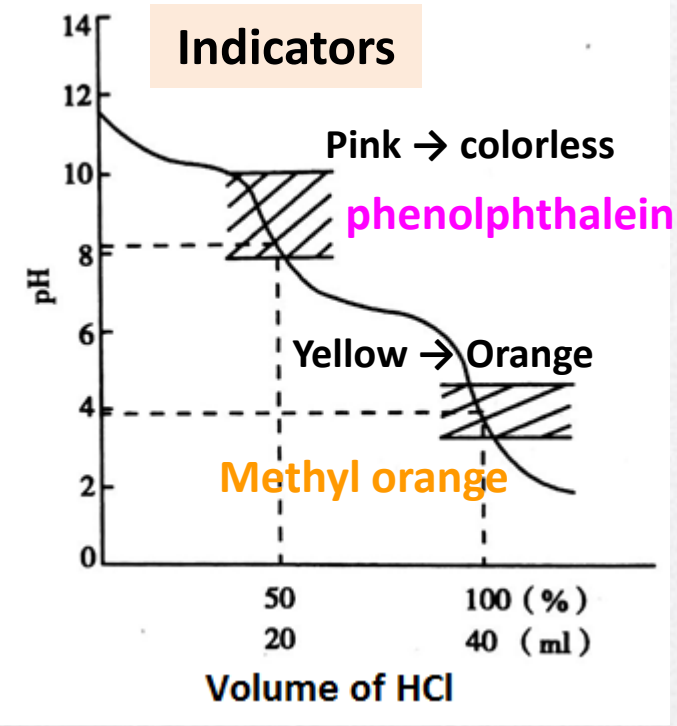
pH_{sp1} depends on the dissociation of HCO_3^-.

$$pOH_{sp_1} \approx \frac{pK_{b_1} + pK_{b_2}}{2} = 5.66$$

$$pH_{sp_1} = 14 - pOH_{sp_1} = 8.34$$

4.3 Polyprotic acids and bases titration

HCl (0.10 mol/L) → Na_2CO_3 (0.10 mol/L, 20.00 ml)



Step 2: $HCl + NaHCO_3 \rightarrow H_2CO_3 + NaCl$

$H_2CO_3 \rightarrow H_2O + CO_2\uparrow$

pH_{sp2} depends on H_2CO_3.

$$pH_{sp_2} = -\lg[H^+] = -\lg\sqrt{c_a K_{a_1}}$$

$$= -\lg\sqrt{0.04 \times 4.2 \times 10^{-7}} = 3.78$$

Practice:

Can 0.1 mol/L oxalic acid ($H_2C_2O_4$) be titrated with 0.1mol/L NaOH accurately or not? If can, how many end point break can be observed? Explain why. $K_{a_1} = 5.6 \times 10^{-2}$; $K_{a_2} = 1.5 \times 10^{-4}$.